



EDUDOC - AN AUTOMATED FRAMEWORK FOR ACADEMIC DOCUMENT COMPLIANCE AND EVALUATION

A MAIN PROJECT REPORT

Submitted by

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MLM24MCA- 2015

To

the APJ Abdul Kalam Technological University

in Partial fulfillment of the requirements for

the award of the degree

Of

MASTER OF COMPUTER APPLICATIONS

Under the Guidance of

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DEPARTMENT OF COMPUTER APPLICATIONS
MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

(Affiliated to APJ Abdul Kalam Technological University)

APRIL 2026



MANGALAM COLLEGE OF ENGINEERING
Accredited by NAAC & ISO 9001:2000 Certified Institution
DEPARTMENT OF COMPUTER APPLICATIONS

VISION

To become a centre of excellence in computer applications, competent in the global ecosystem with technical knowledge, innovation with a sense of social commitment.

MISSION

- To serve with state of the art education, foster advanced research and cultivate innovation in the field of computer applications.
- To prepare learners with knowledge skills and critical thinking to excel in the technological landscape and contribute positively to society.

Program Educational Objectives

- PEO I : Graduates will possess a solid foundation and in-depth understanding of computer applications and will be equipped to analyze real-world problems, design and create innovative solutions, and effectively manage and maintain these solutions in their professional careers.
- PEO II: Graduates will acquire technological advancements through continued education, lifelong learning and research, thereby making meaningful contributions to the field of computing.
- PEO III: Graduates will cultivate team spirit, leadership, communication skills, ethics, and social values, enabling them to apply their understanding of the societal impacts of computer applications effectively.

Program Specific Outcomes

- PSO I: Apply advanced technologies through innovations to enhance the efficiency of design and development.
- PSO II: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

MAPPING OF PO-PSO-SDG

1. MAPPING WITH PROGRAM OUTCOMES (PO): -

SL.NO	POs ADDRESSED	RELEVANCE TO PROJECT
1	PO1	Applied by using programming logic and coding fundamentals to develop the system and implement automated document compliance checking and evaluation.
2	PO2	In EduDOC, problem analysis is applied by identifying issues in manual academic document evaluation and designing a system that automatically checks compliance with formatting and submission guidelines.
3	PO3	EduDOC provides an innovative solution by developing an automated framework that evaluates academic documents, ensuring consistency, transparency, and efficiency in academic assessment.
4	PO4	Utilized modern tools such as Django, MySQL, and document processing libraries to implement evaluation algorithms.
5	PO6	EduDOC is developed using structured planning, phased module implementation, and systematic resource management to ensure efficient project execution and deployment.
4	PO8	The development of EduDOC required learning and adapting to modern technologies such as document processing libraries, demonstrating continuous skill development and technology adoption.

LIST OF PROGRAM OUTCOMES (POs):

PO1 (Foundation Knowledge): Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving.

PO2 (Problem Analysis): Identify, review, formulate and analyse problems for primarily focussing on customer requirements using critical thinking frameworks.

PO3 (Development of Solutions): Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.

PO4 (Modern Tool Usage): Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.

PO5 (Individual and Teamwork): Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.

PO6 (Project Management and Finance): Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management.

PO7 (Ethics): Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware.

PO8 (Life-long learning): Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

2. MAPPING WITH PROGRAM SPECIFIC OUTCOMES (PSOs):-

SL.NO	PSOs ADDRESSED	RELEVANCE TO PROJECT
1	PSO 1	Used advanced technologies such as Python, Django framework, MySQL database, and document processing libraries (python-docx, PyPDF).
2	PSO 2	The project applies computing principles including database design, algorithmic rule validation, and system architecture

LIST OF PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO 1: Apply advanced technologies through innovations to enhance the efficiency of design development.

PSO 2: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

3. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG NO	SDGs ADDRESSED	RELEVANCE TO PROJECT
SDG 4	Quality Education	Improves academic evaluation by ensuring standardized guideline compliance and structured feedback for students.
SDG 9	Industry, Innovation, and Infrastructure	Introduces an innovative digital system for automated academic document analysis and evaluation.

SUSTAINABLE DEVELOPMENT GOALS (SDGs):

SDG 1 – No Poverty-End poverty in all its forms everywhere.

SDG 2 – Zero Hunger-End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG 3 – Good Health and Well-Being-Ensure healthy lives and promote well-being for all at all ages.

SDG 4 – Quality Education-Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 5 – Gender Equality-Achieve gender equality and empower all women and girls.

SDG 6 – Clean Water and Sanitation-Ensure availability and sustainable management of water and sanitation for all.

SDG 7 – Affordable and Clean Energy-Ensure access to affordable, reliable, sustainable, and modern energy for all.

SDG 8 – Decent Work and Economic Growth-Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

SDG 9 – Industry, Innovation, and Infrastructure-Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

SDG 10 – Reduced Inequality-Reduce inequality within and among countries.

SDG 11 – Sustainable Cities and Communities-Make cities and human settlements inclusive, safe, resilient, and sustainable.

SDG 12 – Responsible Consumption and Production-Ensure sustainable consumption and production patterns.

SDG 13 – Climate Action-Take urgent action to combat climate change and its impacts.

SDG 14 – Life Below Water-Conserve and sustainably use the oceans, seas, and marine resources.

SDG 15 – Life on Land -Protect, restore, and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification, halt and reverse land degradation, and halt biodiversity loss.

SDG 16 – Peace, Justice, and Strong Institutions- Promote peaceful and inclusive societies, provide access to justice for all, and build effective, accountable, and inclusive institutions.

SDG 17 – Partnerships for the Goals -Strengthen the means of implementation and revitalize the global partnership for sustainable development.

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

DEPARTMENT OF COMPUTER APPLICATIONS

APRIL 2026



DECLARATION

*I hereby certify that the work which is being presented in the project entitled “EDUDOC - AN AUTOMATED FRAMEWORK FOR ACADEMIC DOCUMENT COMPLIANCE AND EVALUATION” submitted in the **DEPARTMENT OF COMPUTER APPLICATIONS** is an authentic record of my own work carried under the supervision of **Ms. ANJU JOHN, ASSISTANT PROFESSOR**. This study has not been submitted to any other institution or university for the award of any other degree. This report has been checked for plagiarism by the college and the similarity index is within permissible limits set by the college.*

Name & Signature of Student

Date:

Place:

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR
DEPARTMENT OF COMPUTER APPLICATIONS

APRIL 2026



CERTIFICATE

*Certified that the report titled “**EduDOC An Automated Framework for Academic Document Compliance and Evaluation**” submitted by “**Aparna Jiji MLM24MCA-2015**” to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the **DEGREE OF MASTER OF COMPUTER APPLICATIONS** by **APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY** is a bonafide record of the MAIN PROJECT work carried out by her under our guidance and supervision during the academic year 2025-26.*

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ACKNOWLEDGEMENT

I am greatly indebted to the authorities of Mangalam College of Engineering for providing the necessary facilities to successfully complete my Project on the topic “EduDOC An Automated Framework for Academic Document Compliance and Evaluation”

I express my sincere thanks to **Dr. Vinodh P Vijayan**, Principal, Mangalam College of Engineering for providing the facilities to complete my Project successfully.

I thank and express my solicit gratitude to **Ms. Divya S B**, HOD, Department of Computer Applications, Mangalam College of Engineering, for her invaluable help and support which helped me a lot in successfully completing this Project work.

I express my gratitude to my Internal Guide, **Ms. Anju John**, Assistant professor, Department of Computer Applications for the suggestions and encouragement which helped in the successful completion of my Project.

Furthermore, I would like to acknowledge with much appreciation the crucial role of the faculties especially Project coordinator, **Ms. Banu Sumayya S**, Department of Computer Applications, Mangalam College of Engineering, who gave the permission to use all the required equipment and the necessary resources to complete the presentation & report.

Finally, I would like to express my heartfelt thanks to my parents who were very supportive both financially and mentally and for their encouragement to achieve my goal.

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ABSTRACT

Academic institutions require students to follow strict formatting, structural, and submission guidelines when preparing academic documents such as assignments, reports, and project submissions. However, manual verification of these guidelines by instructors is time-consuming, inconsistent, and prone to oversight. This project proposes EduDOC (An Automated Framework for Academic Document Compliance and Evaluation), a web-based system designed to automate the verification and evaluation of academic documents based on predefined institutional guidelines.

The system enables teachers to configure classroom-specific compliance rules such as page count, font type, font size, line spacing, margins, and document structure. Students submit their documents through the platform, after which the system automatically analyzes the files to check formatting compliance, detect plagiarism within the classroom corpus, and identify spelling inconsistencies. A rule-based evaluation engine processes the document and generates a structured compliance report highlighting violations and providing detailed feedback. Annotated versions of submitted documents are also generated to visually indicate issues.

EduDOC improves transparency and efficiency in academic document assessment by providing teachers with automated evaluation support and detailed submission analytics, while allowing students to clearly understand guideline violations and improve their document preparation skills. The system follows a role-based architecture with Admin, Teacher, and Student modules, ensuring controlled access, audit logging, and systematic academic workflow management.

Overall, the proposed framework enhances the reliability, scalability, and fairness of academic document evaluation while reducing the manual workload of instructors.

Keywords:

Academic Compliance, Document Evaluation, Plagiarism Detection, Formatting Verification, Rule-Based Assessment, Submission Management, Educational Technology.

Mapping with Sustainable Development Goals	SDG 4 - Quality Education SDG 9 - Industry, Innovation, and Infrastructure
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CONTENTS

CONTENTS	PAGE NO.
ACKNOWLEDGEMENT	I
ABSTRACT	II
LIST OF FIGURES	VI
LIST OF ABBREVIATIONS	VII
Chapter 1: INTRODUCTION	1
1.1 Background	1
1.2 Motivation	2
1.3 Scope and Objectives	2
1.4 Problem Statement	5
Chapter 2: REQUIREMENTS AND ANALYSIS	6
2.1 Existing System and Problem Statement	6
2.2 Solution Overview	7
2.3 Feasibility Study	9
Chapter 3 LITERATURE REVIEW	11
3.1 Automated Formatting Verification Using Gradient Boosting [Nasyrov et al. (2020)]	11
3.2 Automated Document Formatting Using Python [Venkatesan & Hariharan (2021)]	11
3.3 Document Processing and Workflow Automation [Zahorodko & Merzlykin (2021)]	12
3.4 Word Document Parsing Using Python [Kong et al. (2021)]	13
3.5 Automated Standard Control of Academic Documents [Kozyreva et al. (2021)]	13
3.6 Parsing Electronic Text Documents for Compliance Checking [Martsinkevich et al. (2022)]	14
3.7 NLP-Based Text Extraction and Analysis [Sefara et al. (2022)]	15
3.8 Automated Plagiarism Detection in Academic Documents [Sharyhin et al. (2025)]	15

3.9	AI-Based Document Verification Systems [He et al. (2025)]	16
3.10	Hybrid Plagiarism Detection Using Lexical and Semantic Analysis [Recent Research (2026)]	17
Chapter 4	DIAGRAMS	18
4.1	Data Flow Diagram (DFD)	18
	4.1.1 DFD Level 0	18
	4.1.2 DFD Level 1	19
4.2	Use Case Diagram	20
4.3	Class diagram	21
4.4	State Diagram	22
4.5	Activity Diagram	22
4.6	Sequence Diagram	23
Chapter 5:	METHODOLOGY	25
5.1	Module Descriptions	25
	5.1.1 Authentication and User Management Module	25
	5.1.2 Admin Management Module	27
	5.1.3 Teacher Classroom Management Module	28
	5.1.4 Student Submission Module	29
	5.1.5 Report Generation Module	30
5.2	Algorithms	31
	5.2.1 Document Parsing Algorithm	31
	5.2.2 Formatting Compliance Verification Algorithm	32
	5.2.3 Spelling Detection Algorithm	33
	5.2.4 Evaluation Report Generation Algorithm	34
5.3	System Architecture	35
	5.3.1 Presentation Layer	36
	5.3.2 Application Layer	36
	5.3.3 Data Layer	37

	5.3.4 Workflow of the System	38
	5.3.5 Advantages of the Architecture	38
5.4	Case Study	39
	5.4.1 Teacher Registration and Approval	39
	5.4.2 Student Account Creation	39
	5.4.3 Classroom Creation by Teacher	40
	5.4.4 Student Joining Classroom and Submitting Document	40
	5.4.5 Automated Document Evaluation	40
	5.4.6 Report Generation and Feedback	41
	5.4.7 Outcome of the Case Study	41
Chapter 6	IMPLEMENTATION DETAILS	42
6.1	Implementation Procedures	42
6.2	Testing Methods and Results	44
Chapter 7	ADVANTAGES AND DISADVANTAGES	47
7.1	Advantages	47
7.2	Disadvantages	48
Chapter 8	RESULT & DISCUSSION	49
8.1	Analysis	49
8.2	Results	50
8.3	Discussions	50
Chapter 9	CONCLUSION	52
9.1	Summary	52
9.2	Limitations	53
9.3	Future Scope	54
9.4	Applications	54
Chapter 10	Appendices	56
Chapter 11	References	60

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
4.1	DFD Diagram Level 0	18
4.2	DFD Diagram Level 1 - Admin	19
4.3	DFD Diagram Level 1- Teacher	19
4.4	DFD Diagram Level 1 - Student	20
4.5	UseCase Diagram	20
4.6	Class Diagram	21
4.7	State Diagram	22
4.8	Activity Diagram	23
4.7	Sequence Diagram	24
5.1	Teacher Registration	26
5.2	Login Page	26
5.3	Admin Dashboard	27
5.4	Teacher Dashboard	28
5.5	Student Submission	29
5.6	Document Evaluation	30
5.7	System Architecture	35

LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	- Artificial Intelligence
API	- Application Programming Interface
CSV	- Comma Separated Values
DOC/DOCX	- Microsoft Word Document Format
EduDOC	- Educational Document Compliance System
HTML	- HyperText Markup Language
JSON	- JavaScript Object Notation
KPI	- Key Performance Indicator
PDF	- Portable Document Format
SDG	- Sustainable Development Goal
SQL	- Structured Query Language
UI	- User Interface
URL	- Uniform Resource Locator
XLSX	- Microsoft Excel Spreadsheet Format

CHAPTER 1

INTRODUCTION

1.1 Background

In the academic environment, written documents such as assignments, project reports, research papers, and dissertations play a crucial role in evaluating a student's understanding of a subject. Educational institutions generally provide detailed formatting and structural guidelines that students must follow while preparing academic documents. These guidelines often include specifications related to page layout, margins, font type, font size, line spacing, headings, citation formats, and overall document organization. Adhering to these standards ensures uniformity, readability, and professional presentation of academic work.

Despite the availability of such guidelines, many students struggle to follow them accurately. Academic documents are usually prepared using word processing software, and minor formatting inconsistencies can easily occur during editing, copying content from multiple sources, or converting files between formats. As a result, instructors frequently encounter submissions that deviate from the required formatting standards. Checking these documents manually for compliance with institutional guidelines becomes a time-consuming and repetitive task, especially when dealing with a large number of submissions.

In addition to formatting issues, academic integrity has become a growing concern in educational institutions. The widespread availability of digital resources has made it easier for students to access and reuse content from various sources. While referencing and citation are legitimate academic practices, improper use of external material may lead to plagiarism. Detecting such issues manually can be difficult and inefficient, particularly when instructors must review multiple documents within limited time frames.

Recent advancements in document processing, natural language processing, and automated evaluation techniques have made it possible to develop systems that assist educators in analyzing academic documents automatically. Automated systems can examine document structure, detect formatting inconsistencies, analyze textual content, and generate feedback reports for students. Such systems help reduce the workload of educators while ensuring consistent and objective evaluation of submitted documents.

The EduDOC system, titled “An Automated Framework for Academic Document Compliance and Evaluation,” is designed to address these challenges. The system provides a structured environment where academic documents can be submitted, analyzed, and evaluated automatically according to

predefined institutional guidelines. By combining rule-based document analysis with automated reporting mechanisms, EduDOC aims to streamline the process of academic document verification and improve the overall quality of submissions.

1.2 Motivation

Academic institutions increasingly rely on digital platforms for assignment submission and evaluation. While online learning management systems allow students to upload their work electronically, the responsibility of verifying formatting compliance and academic integrity still largely falls on instructors. Reviewing each document manually to check formatting guidelines, structural consistency, and potential plagiarism requires significant time and effort. This situation becomes even more challenging in courses with large numbers of students.

Another issue observed in academic environments is the lack of immediate feedback for students regarding formatting errors. In many cases, students become aware of formatting mistakes only after their documents have been evaluated and returned by the instructor. The development of EduDOC is motivated by the need to simplify and standardize the evaluation of academic documents. By automating repetitive verification tasks, the system allows educators to focus more on the conceptual and analytical aspects of student work rather than spending excessive time checking formatting details. At the same time, students benefit from receiving structured feedback that helps them understand and correct formatting or structural errors in their documents.

Another motivating factor is the growing interest in intelligent educational technologies that enhance the learning experience. Integrating automated document analysis with submission management provides a practical example of how software systems can support teaching and learning activities. The EduDOC framework aims to demonstrate how automation can be applied effectively in academic environments while maintaining transparency and accountability in the evaluation process.

1.3 Scope and Objectives

Scope of the System

The EduDOC system is designed to operate as a web-based platform that assists educational institutions in verifying and evaluating academic documents submitted by students. The system focuses on the analysis of structured digital documents, particularly those created using widely used word processing formats. By examining various formatting and structural attributes of a document, the system determines whether the submission adheres to predefined academic guidelines.

In many academic settings, instructors are required to review a large number of student submissions while ensuring that each document follows institutional formatting standards. The scope of EduDOC includes providing automated support for this process by analyzing document layout, structure, and formatting properties. The system is capable of examining document elements such as fonts, paragraph spacing, margins, headings, and other structural components that contribute to the overall presentation of academic work. By performing these checks automatically, the platform reduces the time and effort required for manual verification.

The scope of the system includes automated compliance verification, document analysis, plagiarism detection support, and generation of detailed feedback reports. The platform also provides role-based access for administrators, teachers, and students, allowing each user group to interact with the system according to their responsibilities.

Teachers can create classrooms, define submission rules, and review evaluation reports generated by the system. Students can upload their documents and receive structured feedback regarding compliance with the specified guidelines. Administrators manage user accounts and monitor system activity to ensure smooth operation of the platform.

The system also maintains records of document submissions and evaluation results, which can assist instructors in tracking student performance and submission history. This functionality helps ensure transparency in the evaluation process and enables better management of academic activities related to assignments and document submissions.

Objectives of the Project

The primary objective of the EduDOC system is to develop a comprehensive automated framework capable of evaluating academic documents according to predefined formatting and structural guidelines. Academic institutions often establish detailed rules regarding the presentation and structure of reports, assignments, and research documents. Ensuring that these rules are followed consistently can be challenging when evaluation is performed manually. The EduDOC system addresses this challenge by introducing an automated mechanism that analyzes submitted documents and verifies whether they comply with the specified academic standards.

The system is designed to support both educators and students by providing a reliable, transparent, and efficient platform for document submission and evaluation. By integrating automated document analysis with structured reporting, the platform aims to simplify the verification process, reduce manual workload, and improve the overall quality of academic submissions.

The specific objectives of the project are outlined as follows:

- **To design and develop a web-based system for automated academic document compliance-checking:** The system should provide a user-friendly interface through which students can upload their academic documents and instructors can manage submission rules, monitor student submissions, and review evaluation results. The web-based architecture ensures accessibility and ease of integration within institutional environments.
- **To implement rule-based analysis mechanisms that verify formatting elements such as font style, font size, margins, line spacing, and page structure:** The system will analyze the structural properties of submitted documents and compare them with predefined guidelines. This rule-driven approach allows the system to automatically detect formatting inconsistencies and deviations from institutional standards.
- **To enable teachers to define document guidelines that can be automatically applied during evaluation:** Teachers should be able to configure document requirements for each assignment or classroom. These requirements may include formatting specifications, document length constraints, or structural expectations. Once defined, the system applies these rules automatically during the evaluation process.
- **To provide a structured report highlighting compliance results and detected issues within submitted documents:** After analyzing a document, the system should generate a detailed report summarizing compliance status, identified violations, and relevant feedback. This report helps instructors quickly understand the quality of the submission and enables students to recognize and correct formatting errors.
- **To support plagiarism detection mechanisms and document similarity analysis where applicable:** The system should include mechanisms that assist in identifying potential similarities between submitted documents or previously stored submissions. Such functionality helps maintain academic integrity by discouraging unauthorized reuse of content.
- **To reduce the time required for manual evaluation of document formatting and structure:** By automating repetitive verification tasks, the system minimizes the effort required from instructors when reviewing a large number of submissions. This allows educators to focus more on evaluating the intellectual content and analytical quality of student work.

- **To improve the quality of academic submissions by providing students with feedback:**

The feedback generated by the system highlights specific formatting errors or structural issues within the document. This guidance helps students understand how their documents deviate from the expected standards and encourages them to improve their document preparation practices.

1.4 Problem Statement

Educational institutions frequently require students to submit assignments, reports, and research documents that follow specific formatting and structural guidelines. Ensuring that these guidelines are correctly followed is essential for maintaining uniformity and academic standards. However, verifying compliance with such guidelines is often performed manually by instructors, which can be inefficient and time-consuming.

Manual verification involves examining each document for formatting issues such as incorrect margins, inconsistent font styles, improper line spacing, missing headings, and deviations from prescribed document structures. When instructors must evaluate large numbers of submissions, this process becomes repetitive and prone to oversight. Additionally, manual evaluation may result in inconsistencies when different instructors interpret guidelines differently.

Another challenge arises from the increasing availability of digital content and the ease with which information can be copied from external sources. Detecting potential plagiarism or identifying reused content requires careful analysis and specialized tools. Without automated support, instructors may find it difficult to identify such issues accurately.

Students also face difficulties when preparing academic documents according to strict guidelines. Minor formatting mistakes may go unnoticed during document preparation, and students often receive feedback only after the evaluation process is completed. This limits their ability to correct errors before submission and may negatively affect the quality of their work.

These challenges highlight the need for a system that can automatically analyze academic documents and verify their compliance with institutional guidelines. Such a system should be capable of processing submitted documents, identifying formatting inconsistencies, analyzing document structure, and generating clear evaluation reports. By automating these processes, the workload of instructors can be significantly reduced while ensuring that evaluation criteria are applied consistently. The EduDOC system addresses this problem by providing an integrated platform for automated document compliance checking and evaluation. Through rule-based analysis and structured reporting, the system supports educators in managing document submissions efficiently while helping students improve the quality and accuracy of their academic work.

CHAPTER 2

REQUIREMENTS AND ANALYSIS

2.1 Existing System and Problem Statement

In contemporary academic environments, students are frequently required to submit assignments, project reports, research papers, and other forms of written documentation as part of their academic assessment. These documents are expected to follow specific formatting and structural guidelines defined by educational institutions. Such guidelines ensure uniformity, clarity, and professionalism in academic submissions. Typically, these guidelines include requirements related to page layout, margin settings, font styles, font sizes, line spacing, heading hierarchy, citation formats, and overall document organization.

Despite the availability of clearly defined guidelines, many academic institutions continue to rely on manual processes for verifying whether submitted documents comply with these requirements. Students prepare their documents using word processing tools such as Microsoft Word or similar applications and submit them through institutional portals, email systems, or learning management systems. Once submitted, instructors must manually review each document to determine whether the formatting and structural requirements have been properly followed.

Manual verification of document formatting can be tedious and time-consuming, especially when instructors are responsible for evaluating submissions from large numbers of students. Each document must be opened and carefully inspected to check elements such as margin sizes, font properties, paragraph spacing, alignment, and section headings. Even minor inconsistencies in formatting can require additional time to detect and verify. As the number of submissions increases, the effort required for manual verification grows significantly.

Another limitation of the existing system is the potential for human error during the evaluation process. When instructors review documents manually, there is a possibility that certain formatting violations may be overlooked. In addition, different instructors may interpret formatting guidelines differently, which can lead to inconsistencies in the evaluation process. Such variations can affect the fairness and reliability of document assessment.

Academic integrity also presents a major challenge within the current document evaluation process. With easy access to online resources and digital publications, students may unintentionally include external content without proper citation. Detecting potential plagiarism manually requires careful comparison with existing sources and previously submitted documents, which can be difficult

without specialized tools. Although external plagiarism detection tools are available, they are often separate from the document submission workflow and may not provide integrated support for formatting verification.

Students themselves often encounter difficulties while attempting to follow complex formatting guidelines. Academic formatting standards can be detailed and sometimes difficult to interpret, particularly for students who are unfamiliar with professional document preparation practices. During the editing process, formatting may change unintentionally due to copying text from other documents, inserting images or tables, or modifying document styles. As a result, students may unknowingly submit documents that contain formatting errors.

In many cases, students only receive feedback regarding formatting issues after their documents have been reviewed and graded. This delayed feedback reduces the opportunity for students to correct errors before final submission. Consequently, students may lose marks due to formatting mistakes that could otherwise have been corrected earlier.

In addition to these issues, the existing document evaluation process often lacks transparency and structured reporting. Instructors may provide feedback in different formats or levels of detail, making it difficult for students to clearly understand which formatting rules were violated and how they can correct them. A lack of standardized evaluation reports can therefore hinder effective communication between instructors and students.

The primary problems associated with the existing system can therefore be summarized as follows:

- Dependence on manual verification of formatting and structural rules
- Significant time and effort required from instructors when evaluating large numbers of documents
- Possibility of human error during manual document inspection
- Inconsistent interpretation of guidelines across different evaluators
- Limited integration of plagiarism detection within the document evaluation workflow
- Delayed and sometimes unclear feedback provided to students

2.2 Solution Overview

To overcome the limitations identified in the existing system, the EduDOC framework has been developed as an integrated solution for automated academic document compliance and evaluation. The system is designed to assist educational institutions in verifying whether submitted documents

adhere to predefined formatting guidelines and structural standards. By incorporating automated document analysis and rule-based evaluation mechanisms, EduDOC provides a systematic and efficient approach to document verification.

The EduDOC system operates as a web-based platform that supports document submission, guideline management, automated evaluation, and report generation. The platform is accessible through a web browser, allowing students, teachers, and administrators to interact with the system from different locations without requiring specialized software installations.

The core concept behind EduDOC is the use of rule-based document analysis. Instructors can define a set of formatting guidelines that describe how academic documents should be structured. These guidelines may include requirements such as specific font types, font sizes, line spacing values, margin dimensions, and document section structures. Once these rules are configured within the system, they are automatically applied during the document evaluation process.

When a student uploads a document, the system processes the file and extracts relevant structural and formatting attributes. The extracted attributes are then compared with the predefined rules provided by the instructor. Any deviation from the specified guidelines is identified and recorded during the evaluation process.

After the analysis is completed, the system generates a structured evaluation report that summarizes the compliance status of the submitted document. The report includes information regarding which formatting rules were satisfied and which rules were violated.

The system also supports additional features that enhance the overall document evaluation process. For example, the platform can perform basic textual analysis tasks such as spelling checks and word count analysis. These features help identify common issues in academic writing and provide additional insights into the quality of submitted documents.

Another important feature of the EduDOC system is its ability to maintain records of document submissions and evaluation results. This functionality allows instructors to track the submission history of students, monitor evaluation outcomes, and maintain transparency in the assessment process.

The automated evaluation process implemented by EduDOC significantly reduces the time required for manual document inspection. Instead of manually reviewing each formatting rule, instructors can rely on the system to perform detailed analysis and present the results in a structured format. This allows educators to focus more on evaluating the intellectual and conceptual quality of student work rather than spending time checking formatting details.

In addition to supporting instructors, the system also provides significant benefits for students. By receiving clear feedback regarding formatting compliance, students can identify errors in their documents and make corrections before final submission. This promotes better learning practices and helps students develop stronger academic writing skills.

2.3 Feasibility Study

Before developing and implementing a software system, it is essential to evaluate whether the proposed solution is feasible within the available technical, financial, and operational constraints. A feasibility study helps determine whether the system can be developed successfully and whether it will provide practical benefits to its intended users. The feasibility of the EduDOC system can be examined through technical feasibility, economic feasibility, and operational feasibility.

Technical Feasibility

Technical feasibility evaluates whether the proposed system can be implemented using available technologies and development resources. The EduDOC system relies on widely adopted web development technologies and document processing libraries that are capable of analyzing structured digital documents.

Modern programming frameworks provide the necessary tools for developing web-based platforms that support user authentication, database management, file uploads, and automated processing tasks. Document processing libraries allow the system to read and analyze word processing files by extracting information related to fonts, paragraph formatting, and document layout. These technologies make it possible to implement automated compliance checking without requiring complex or proprietary software.

The modular architecture of the EduDOC system also enhances its technical feasibility. Different components of the system, such as user management, document processing, evaluation logic, and report generation, are designed as independent modules. This approach simplifies system maintenance and allows developers to update or extend individual components without affecting the entire system.

Additionally, the system can be deployed on standard web servers and database platforms that are commonly used in educational institutions. Since the technologies involved are well-documented and widely supported, the development and deployment of the system are technically achievable.

Economic Feasibility

Economic feasibility considers the financial implications of developing and deploying the system. The EduDOC platform is designed using open-source technologies and freely available

development tools, which significantly reduces the cost of implementation. Most of the software components required for development, such as web frameworks, databases, and document processing libraries, are available under open-source licenses.

Furthermore, the automation of document verification tasks can result in indirect cost savings by reducing the time and effort required from instructors during the evaluation process. By minimizing manual work, educational institutions can improve productivity and allocate resources more efficiently.

Operational Feasibility

Operational feasibility evaluates whether the proposed system can be effectively integrated into the daily activities of its intended users. The EduDOC system is designed with usability and accessibility as key considerations, ensuring that both students and instructors can interact with the platform easily.

The system provides clear interfaces for uploading documents, defining evaluation rules, and reviewing compliance reports. These interfaces are designed to be intuitive so that users with basic computer skills can operate the system without extensive training.

Since many educational institutions already utilize digital platforms for assignment submission and communication, integrating EduDOC into existing workflows is relatively straightforward. The platform complements existing academic processes by adding automated verification capabilities without disrupting established submission procedures.

Additionally, the system supports role-based access control, ensuring that different users can access only the features relevant to their responsibilities. This improves system security and simplifies the overall user experience.

From an operational perspective, the benefits of automated document verification and structured feedback make the system highly practical for academic environments. The ability to reduce manual workload, improve evaluation consistency, and provide faster feedback to students contributes to the overall effectiveness of the system.

CHAPTER 3

LITERATURE REVIEW

3.1 Automated Formatting Verification Using Gradient Boosting [Nasyrov et al. (2020)]

Nasyrov et al. (2020) proposed an automated approach for verifying document formatting using machine learning techniques. The study introduces a formatting verification method that analyzes structural features of digital documents and evaluates them using a gradient boosting decision tree model. The system extracts attributes such as font size, font type, paragraph spacing, and document layout, and uses these features to determine whether a document satisfies predefined formatting rules.

The proposed model significantly reduces the effort required for manual formatting verification and improves accuracy by consistently applying the same evaluation criteria across documents. The authors demonstrate that machine learning models can efficiently classify documents based on formatting compliance, making them suitable for large-scale document verification tasks.

The research highlights the growing importance of automated document verification systems in educational and professional environments. The techniques proposed in the study provide a foundation for systems like EduDOC that perform rule-based analysis of academic documents.

Key Aspects

- Machine learning approach for formatting verification
- Gradient boosting algorithm for classification
- Automated extraction of formatting features
- Detection of formatting inconsistencies
- Reduction of manual document inspection

3.2 Automated Document Formatting Using Python [Venkatesan & Hariharan (2021)]

Venkatesan and Hariharan (2021) introduced an automated framework for document formatting verification using Python-based tools. The study demonstrates how document processing libraries can analyze formatting properties of digital documents and compare them with predefined guidelines. The system reads word processing files and extracts structural attributes such as font style, font size, margin values, paragraph spacing, and alignment. These attributes are evaluated

using a rule-based verification approach. When formatting violations are detected, the system generates a report highlighting the issues.

The framework emphasizes the role of automation in reducing manual effort involved in reviewing academic documents. By programmatically evaluating formatting rules, the system improves consistency and accuracy in document assessment. The study also illustrates how Python libraries can efficiently process document structures and provide detailed feedback on formatting inconsistencies. Such automated verification systems are particularly useful in educational environments where large numbers of assignments or reports must be checked against standardized formatting guidelines.

Key Aspects

- Python-based document analysis system
- Automated extraction of formatting attributes
- Rule-based compliance verification
- Structured error reporting
- Application in academic document evaluation

3.3 Document Processing and Workflow Automation [Zahorodko & Merzlykin (2021)]

Zahorodko and Merzlykin (2021) proposed a method for processing word processing documents to support automated document workflow systems. Their research focuses on analyzing the structural elements of documents created in Microsoft Word and LibreOffice to enable efficient automated document management. The study demonstrates how document parsing techniques can be applied to identify and interpret the internal structure of digital documents.

The system extracts structural components such as headings, paragraphs, tables, and formatting attributes. By analyzing these elements, the system can determine how a document is organized and whether it conforms to predefined layout requirements. This structural analysis allows automated systems to classify, validate, and process documents more efficiently within digital workflows.

The authors emphasize that document parsing technologies play a crucial role in developing intelligent document verification systems. Such techniques enable automated tools to interpret document structures and detect inconsistencies in formatting or layout. These capabilities are particularly valuable in academic and institutional environments where standardized document formats must be maintained.

Key Aspects

- Document parsing techniques
- Extraction of structural document elements
- Automated workflow processing
- Layout and structure verification

3.4 Word Document Parsing Using Python [Kong et al. (2021)]

Kong et al. (2021) investigated methods for parsing Microsoft Word documents using Python-based tools to support automated document analysis applications. The study focuses on extracting textual and formatting information from Word files in order to enable systematic analysis of document structure and layout. By utilizing Python document processing libraries, the system is capable of reading Word document elements and interpreting their structural organization.

The proposed approach identifies document components such as paragraphs, tables, images, and formatting attributes. These components are then used to reconstruct the logical structure of the document, allowing the system to analyze formatting patterns and layout properties. Through this structured representation, automated systems can evaluate whether documents adhere to predefined formatting or structural guidelines.

The research highlights that document parsing plays a fundamental role in the development of automated document verification systems. Accurate extraction of structural elements enables further processes such as rule-based compliance checking, document classification, and automated report generation.

Key Aspects

- Python-based document parsing
- Extraction of structural document elements
- Reconstruction of document layout
- Automated document analysis

3.5 Automated Standard Control of Academic Documents [Kozyreva et al. (2021)]

Kozyreva et al. (2021) proposed a system for automated standard control of academic documents aimed at verifying whether submitted documents comply with institutional formatting requirements.

The research focuses on developing a verification mechanism that systematically analyzes document structure and layout to ensure adherence to predefined academic standards.

The system examines structural elements such as headings, section organization, paragraph arrangement, and overall document layout. These components are compared against predefined formatting rules defined by the institution. Based on this comparison, the system determines whether the document satisfies the required guidelines or contains deviations from the specified standards.

The study emphasizes that automated standard control systems can significantly improve the reliability and consistency of academic document evaluation. By automating the verification process, such systems reduce manual effort and minimize human errors during document review. This approach also enables faster processing of large numbers of academic submissions while maintaining uniform evaluation criteria.

Key Aspects

- Automated standard verification for academic documents
- Structural analysis of document sections
- Rule-based compliance evaluation
- Improved consistency in document assessment

3.6 Parsing Electronic Text Documents for Compliance Checking [Martsinkevich et al. (2022)]

Martsinkevich et al. (2022) examined the application of document parsing techniques for automated compliance verification in electronic text documents. The study focuses on extracting structural attributes from digital documents and evaluating them against predefined rule sets to determine whether the document adheres to established formatting standards.

The proposed system parses document structures and extracts formatting attributes such as paragraph styles, headings, spacing, and layout information. These extracted attributes are then compared with predefined formatting rules and institutional standards. Based on this comparison, the system identifies compliance or detects formatting violations within the document.

The authors highlight that automated parsing techniques significantly enhance the efficiency and accuracy of document verification systems. By automating the extraction and analysis of document structures, such systems reduce the time required for manual review and improve the consistency of formatting validation. This approach is particularly beneficial in academic and professional

environments where large volumes of documents must be checked against standardized formatting requirements.

Key Aspects

- Automated parsing of electronic documents
- Extraction of formatting attributes
- Rule-based compliance verification
- Reduction of manual document evaluation

3.7 NLP-Based Text Extraction and Analysis [Sefara et al. (2022)]

Sefara et al. (2022) introduced a toolkit for text extraction and analysis designed for natural language processing (NLP) tasks. The toolkit enables automated processing of textual content within digital documents and supports the analysis of linguistic and structural characteristics of text.

Using NLP techniques, the system can analyze textual structures, detect spelling errors, and extract meaningful patterns from document content. These capabilities allow automated systems to evaluate written documents and identify issues related to text quality, readability, and linguistic correctness. The toolkit processes document text through a sequence of extraction and analysis stages, enabling efficient interpretation of textual information.

The authors emphasize that NLP techniques can enhance document evaluation systems by providing analytical capabilities beyond basic formatting verification. By incorporating text analysis methods, automated systems can assess the quality of written content and provide more comprehensive feedback. Such approaches are particularly useful in academic environments where both formatting compliance and textual clarity are important for document evaluation.

Key Aspects

- NLP toolkit for text extraction
- Automated linguistic analysis
- Detection of spelling and writing errors
- Application in academic document analysis

3.8 Automated Plagiarism Detection in Academic Documents [Sharyhin et al. (2025)]

Sharyhin et al. (2025) investigated automated methods for detecting plagiarism in academic documents. The study explores techniques for identifying copied or modified content within student

assignments, essays, and research papers. The research focuses on developing automated mechanisms that can analyze textual similarities across multiple documents to detect potential cases of plagiarism.

The proposed system evaluates textual similarity using pattern matching and semantic comparison techniques. By comparing document content with existing text sources, the system can identify duplicated phrases, paraphrased content, and structurally modified sections that may indicate plagiarism. These detection mechanisms help determine whether a document contains content that closely resembles previously submitted or existing documents.

The authors highlight that automated plagiarism detection systems play a crucial role in maintaining academic integrity within educational institutions. Such systems enable instructors and administrators to efficiently analyze large volumes of academic submissions while ensuring fairness and transparency in evaluation. The study also emphasizes that effective plagiarism detection tools must be capable of identifying different forms of plagiarism, including paraphrasing, text substitution, and structural modifications within documents.

Key Aspects

- Automated plagiarism detection system
- Text similarity analysis
- Detection of paraphrased plagiarism
- Support for academic integrity verification

3.9 AI-Based Document Verification Systems [He et al. (2025)]

He et al. (2025) proposed an automated document verification system that integrates artificial intelligence techniques to enhance the accuracy and efficiency of document validation processes. The study focuses on the use of intelligent algorithms to analyze both document structure and textual content in order to determine whether a document complies with predefined standards.

The proposed system examines various document attributes such as structural organization, formatting elements, and textual characteristics. By applying AI-driven analysis methods, the system can automatically detect formatting inconsistencies, structural anomalies, and deviations from expected document layouts. This automated verification process reduces the need for manual inspection while improving the speed and reliability of document evaluation.

The authors demonstrate that AI-based verification systems significantly improve the efficiency of document analysis, particularly when processing large volumes of digital documents. The research

further highlights the importance of integrating artificial intelligence with document processing technologies to develop intelligent document verification systems capable of supporting automated compliance checking and quality assessment.

Key Aspects

- AI-based document verification
- Automated analysis of document structure
- Detection of formatting inconsistencies
- Improved efficiency in document validation

3.10 Hybrid Plagiarism Detection Using Lexical and Semantic Analysis [Recent Research (2026)]

Recent research has explored hybrid approaches to plagiarism detection that combine lexical similarity techniques with semantic analysis methods. A study published in 2026 proposed a system that integrates TF-IDF lexical similarity with Sentence-BERT semantic embeddings to detect plagiarism in academic documents.

The system uses a Support Vector Machine (SVM) classifier to distinguish between original and plagiarized documents. By combining lexical and semantic similarity measures, the system can detect plagiarism even when the original content has been paraphrased or structurally modified.

The study demonstrates that hybrid plagiarism detection techniques are more effective than traditional string-matching approaches. Such techniques are particularly useful for modern academic environments where advanced text modification strategies are commonly used.

Key Aspects

- Hybrid plagiarism detection techniques
- Combination of lexical and semantic similarity analysis
- Machine learning classification using SVM
- Detection of paraphrased plagiarism
- Improved accuracy in academic document analysis

CHAPTER 4

DIAGRAMS

4.1 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data moves through a system. It provides a clear visualization of processes, data stores, data flows, and external entities, making it easier to understand system functionality. DFDs describe what happens to data rather than how it happens, thus abstracting away technical details while focusing on logical flow. Key Components of a DFD:

- **Process:** Represents an operation or activity that transforms input data into output data.
- **Data Flow:** Shows the movement of data between processes, data stores, and entities.
- **Data Store:** Represents places where data is stored temporarily or permanently.
- **External Entity:** Denotes sources or destinations of data outside the system boundary.

4.1.1 DFD Level 0

DFD Level 0 (Context Diagram) represents the EduDOC system as a single process interacting with three external entities: Admin, Teacher, and Student. Students submit documents and receive evaluation reports, teachers manage classrooms and review submissions, and the admin oversees system management such as approvals and batch assignments. The system processes all data flows and generates compliance and evaluation results.

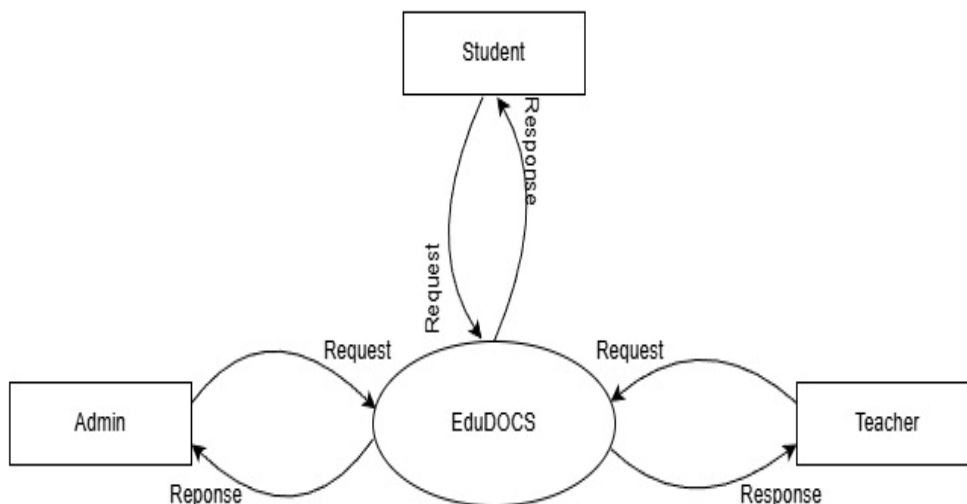


Figure 4.1 DFD Diagram Level 0

4.1.2 DFD Level 1

The Level 1 Data Flow Diagram (DFD) provides a detailed view of the EduDOC system by dividing the main system into internal processes such as user management, classroom management, document submission, document evaluation, and report generation. The system interacts with three external entities: Administrator, Teacher, and Student. The administrator manages users and approves teachers, teachers create classrooms and define document guidelines, and students upload documents for evaluation. The uploaded documents are analyzed by the system to check formatting rules, and a compliance report is generated. All data such as user details, submissions, and reports are stored in the system database.

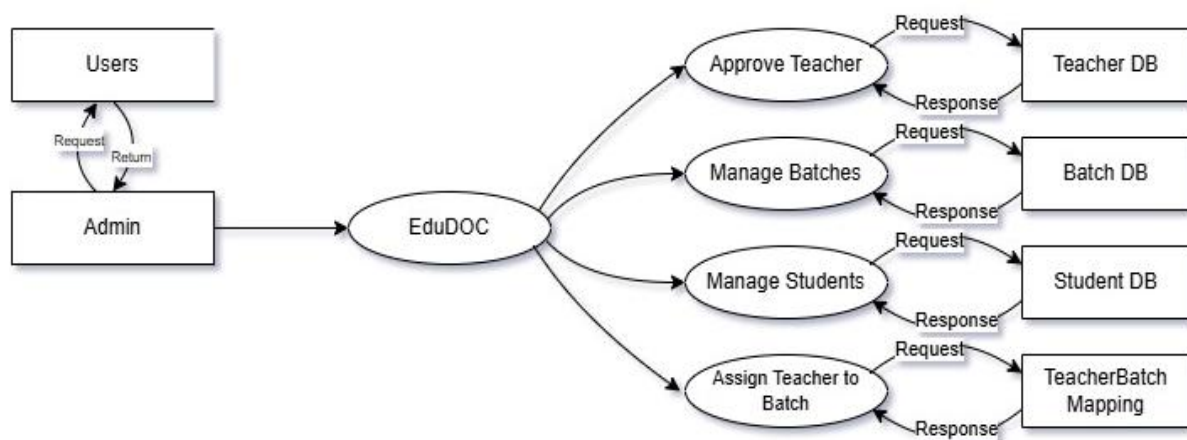


Figure 4.2 DFD Diagram Level 1- Admin

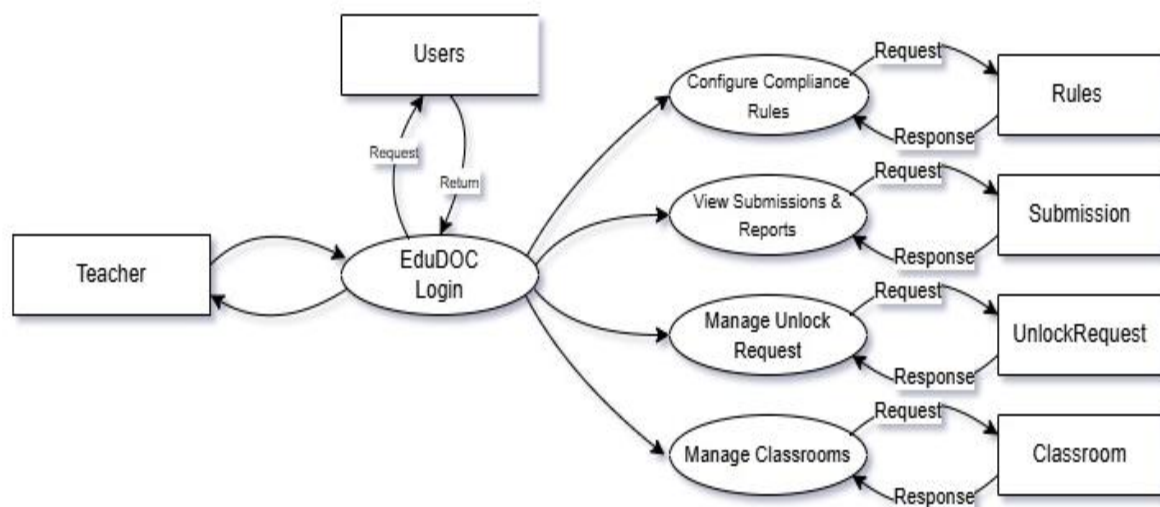


Figure 4.3 DFD Diagram Level 1 – Teacher

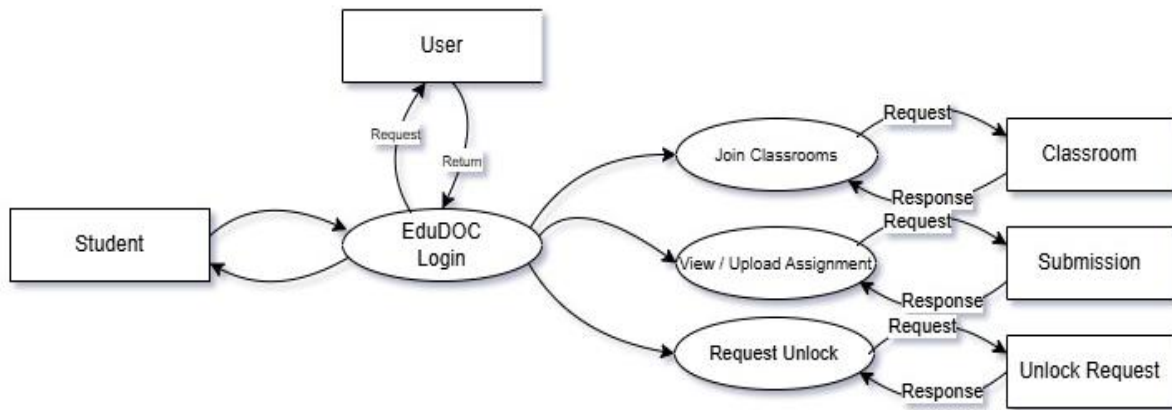


Figure 4.4 DFD Diagram Level 1 - Student

4.2 Use Case Diagram

The Use Case Diagram represents the interaction between the users and the EduDOC system. The system involves three main actors: Administrator, Teacher, and Student. The administrator manages system operations such as approving teacher registrations, creating student accounts, and monitoring user activities. Teachers use the system to create classrooms, define document formatting guidelines, set submission rules, and review student submissions and evaluation reports. Students interact with the system by joining classrooms using a passkey, uploading their academic documents, and viewing the compliance reports generated after document evaluation. The diagram illustrates how these actors communicate with the system to perform different functions related to academic document submission and automated compliance verification.

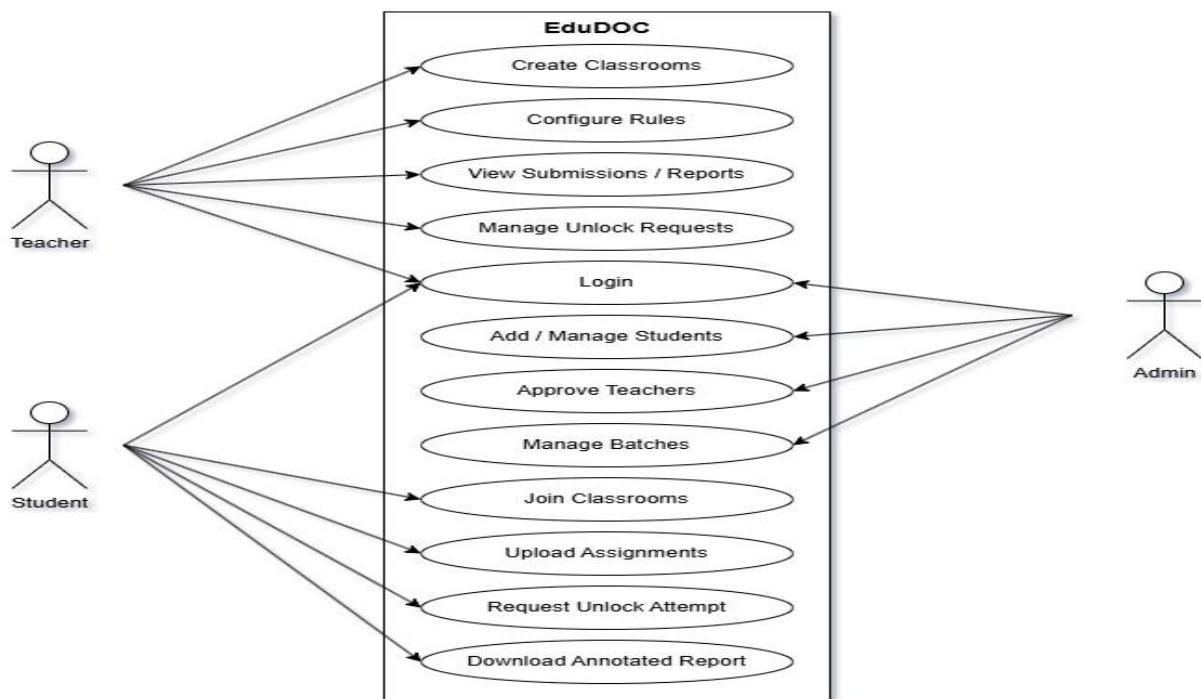


Figure 4.5 UseCase Diagram

Students interact with the system by joining classrooms using a passkey, uploading their academic documents, and viewing the compliance reports generated after document evaluation. The diagram illustrates how these actors communicate with the system to perform different functions related to academic document submission and automated compliance verification.

4.3 Class Diagram

The Class Diagram represents the structural design of the EduDOC system by showing the main classes, their attributes, and the relationships between them. The primary classes in the system include Administrator, Teacher, Student, Classroom, Submission, and Evaluation Report. The Administrator class manages user accounts and system configuration. The Teacher class is responsible for creating classrooms, defining document formatting rules, and monitoring student submissions. The Student class represents users who join classrooms and submit academic documents for evaluation. The Classroom class stores assignment details such as guidelines, submission deadlines, and configuration settings. The Submission class handles document uploads made by students, while the Evaluation Report class stores the results of document analysis and compliance verification. These classes interact with each other to support the processes of classroom management, document submission, automated document evaluation, and report generation within the EduDOC system.

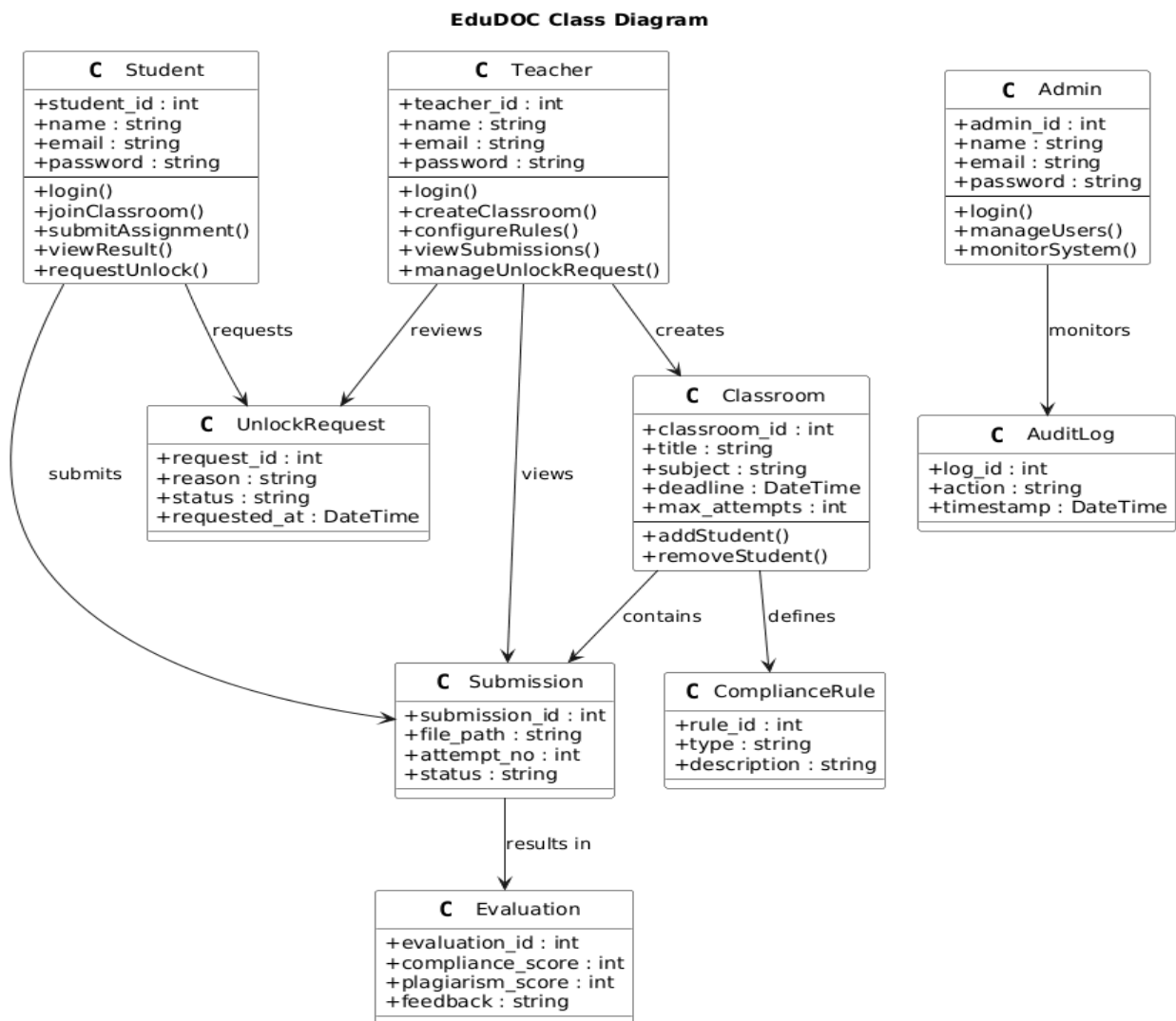


Figure 4.6 Class Diagram

4.4 State Diagram

The State Diagram illustrates the different states through which a document or submission passes during its lifecycle in the EduDOC system. Initially, the system starts in the User Login state, where users authenticate to access the platform. After successful login, teachers may create classrooms and students may join them using a passkey. When a student uploads a document, the submission enters the Document Uploaded state. The system then moves to the Document Evaluation state, where formatting attributes such as font type, size, spacing, and margins are analyzed and compared with predefined rules. Based on the evaluation results, the system generates a Compliance Report state, where the report is stored and displayed to both teachers and students. Finally, the process reaches the Completed state, where the submission and evaluation results remain available for review within the system.

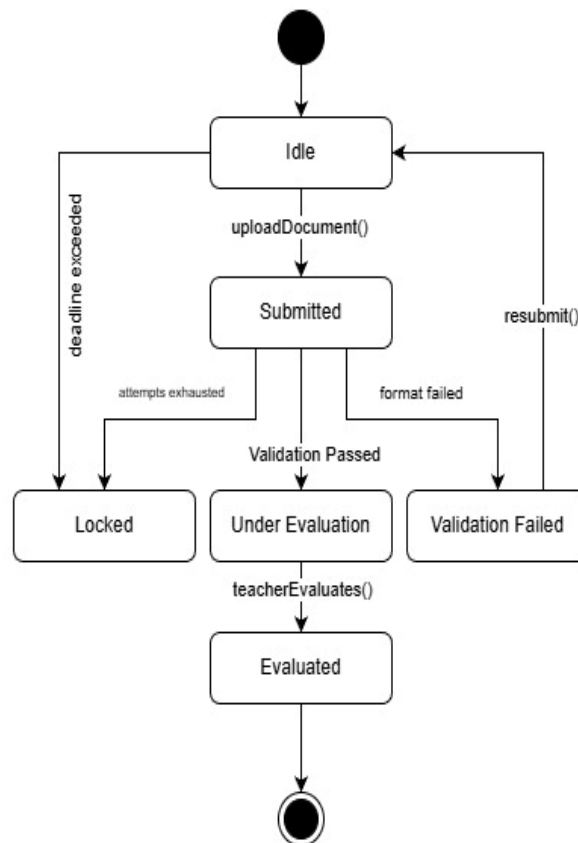


Figure 4.7 State Diagram

4.5 Activity Diagram

The Activity Diagram represents the workflow of operations in the EduDOC system from document submission to evaluation. The process begins when the user logs into the system and accesses the dashboard based on their role. Teachers create classrooms and define document formatting guidelines, while students join classrooms using a passkey. The student then uploads the academic document, which is stored in the system. The document is forwarded to the evaluation process,

where formatting attributes such as font style, size, margins, and spacing are analyzed and compared with predefined rules. After the evaluation is completed, the system generates a compliance report that highlights formatting violations and evaluation results. The report is stored in the database and made available to both teachers and students through their dashboards, completing the workflow.

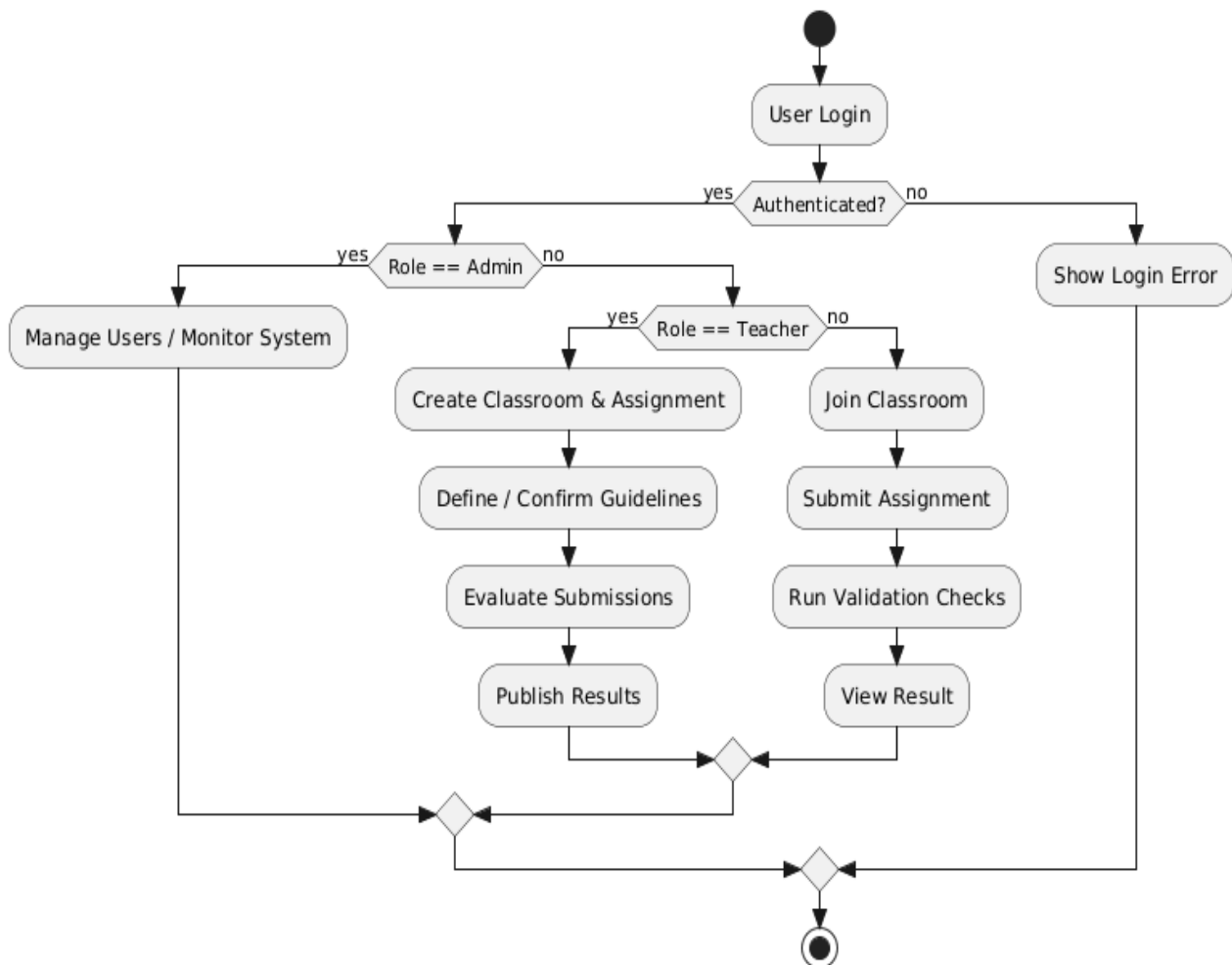


Figure 4.8 Activity Diagram

4.6 Sequence Diagram

The Sequence Diagram illustrates the order of interactions between different components of the EduDOC system during the document submission and evaluation process. The interaction begins when the student logs into the system and uploads an assignment document through the submission interface. The request is sent to the system server, which stores the uploaded file in the database. The system then forwards the document to the evaluation module, where formatting attributes such as font style, font size, spacing, and margins are analyzed and compared with predefined guidelines. After the evaluation is completed, the system generates a compliance report and stores it in the database. Finally, the teacher and student can access the generated report through their respective dashboards to review the evaluation results.

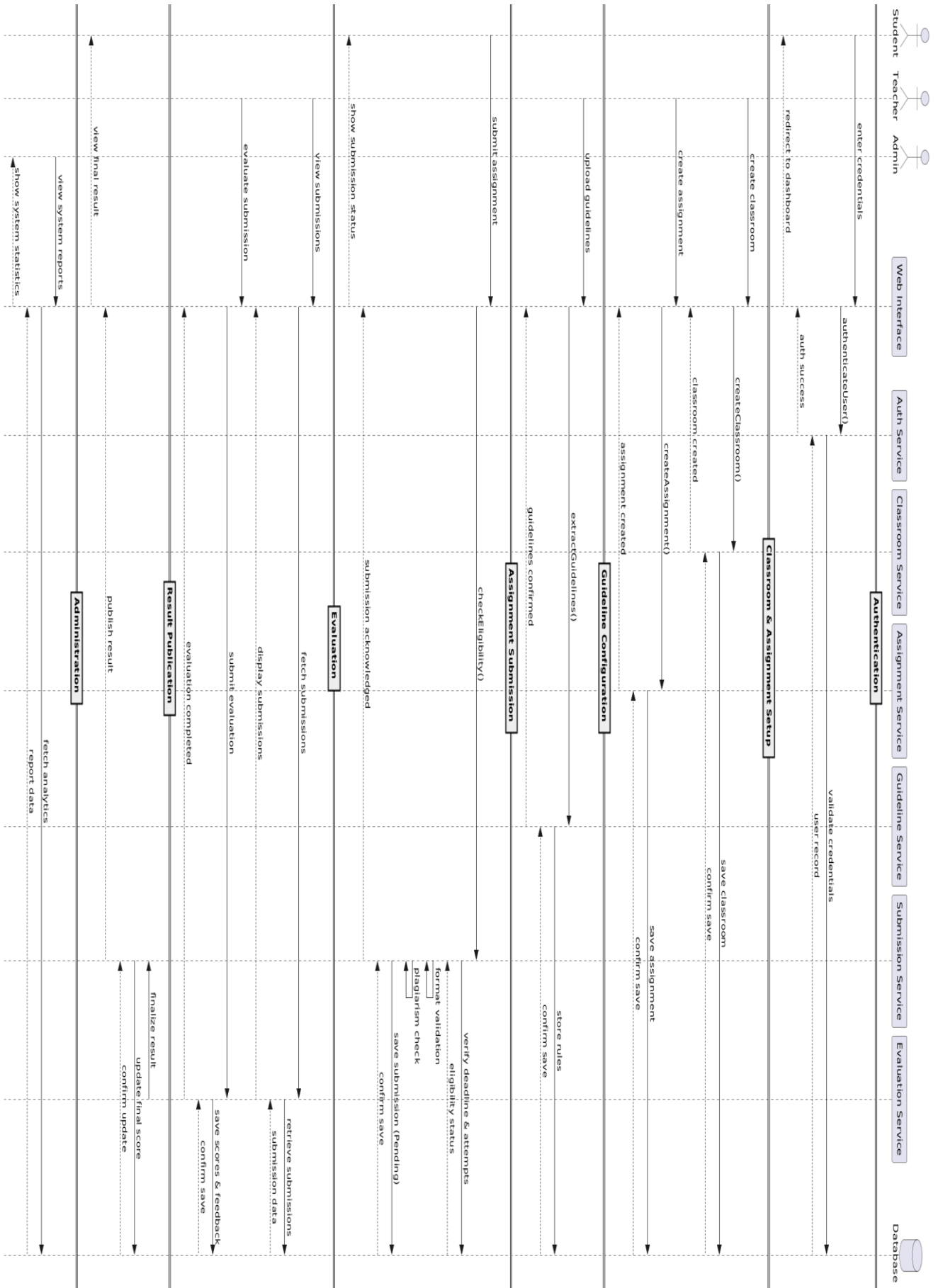


Figure 4.7 Sequence Diagram

CHAPTER 5

METHODOLOGY

5.1 Module Descriptions

The EduDOC system is developed as a modular web-based application that automates the evaluation of academic documents. The system is organized into several functional modules, each responsible for a specific task within the document submission and evaluation workflow. These modules work together to ensure efficient document processing, rule-based compliance verification, plagiarism detection, and report generation.

The modular architecture improves system maintainability, scalability, and reliability. Each module handles a distinct set of operations, allowing the system to process user requests and document evaluations efficiently.

5.1.1 Authentication and User Management Module

The Authentication and User Management module provides the security foundation of the EduDOC system. This module ensures that only authorized users can access the platform and that system functionalities are restricted according to the user's role.

The system supports three primary user roles: Administrator, Teacher, and Student. Each user interacts with the platform through role-based authentication and is granted access only to the features associated with their responsibilities.

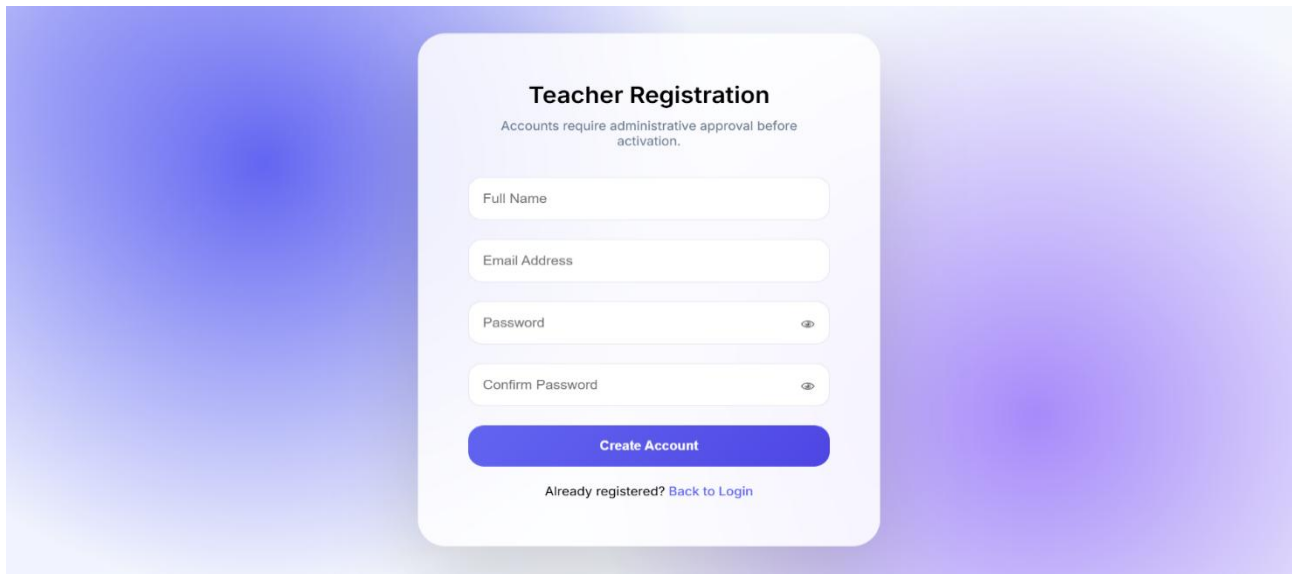
Teachers are allowed to register themselves through the system registration interface. However, newly registered teachers cannot access the system immediately. Their accounts must first be reviewed and approved by the administrator. Only after administrative approval can teachers log in. Student accounts are created by the administrator. The administrator can add students either individually or through bulk upload mechanisms such as CSV or spreadsheet files. Once created, students receive their login credentials and can access the system.

After logging in, students can join classrooms using a passkey provided by the teacher. The authentication module also manages session control and secure credential verification to prevent unauthorized access. Through role-based authentication and administrative approval workflows, the module ensures the security and integrity of the EduDOC platform.

Key functions of this module include:

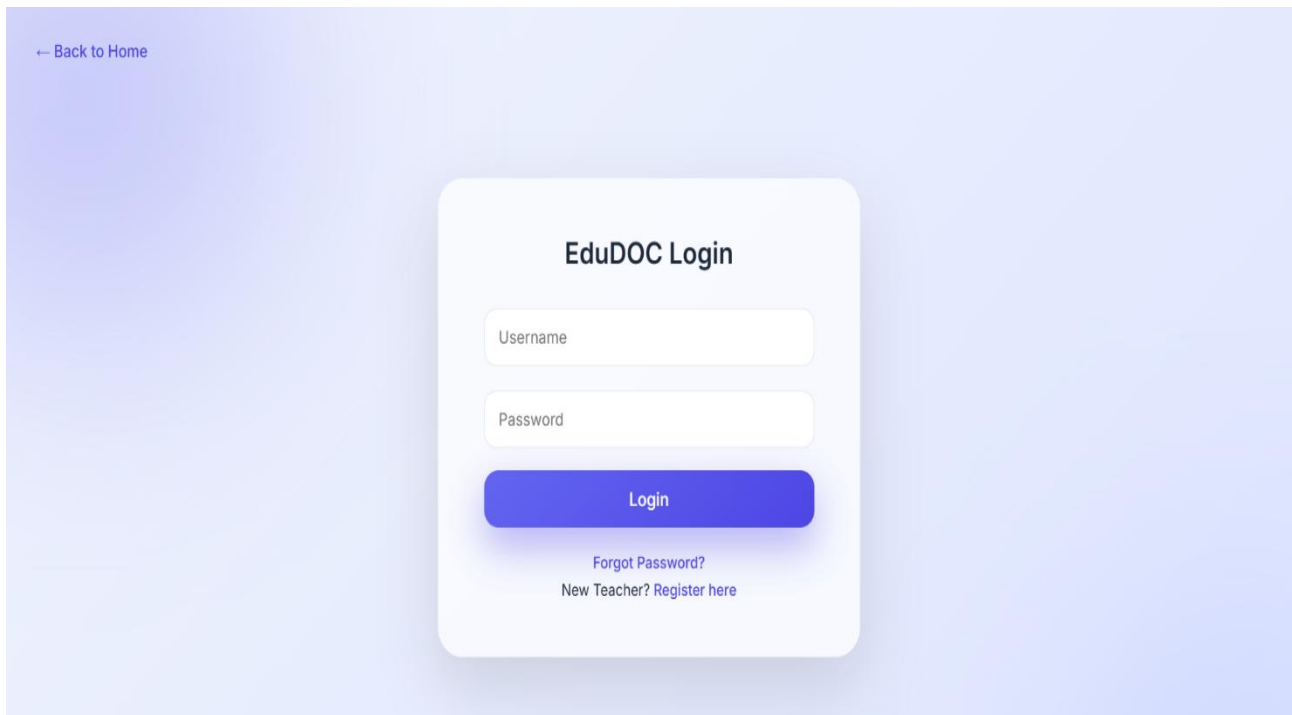
- Teacher self-registration

- Admin approval of teacher accounts
- Admin creation of student accounts
- Secure login authentication
- Role-based access control
- Session and credential management



The image shows a 'Teacher Registration' form centered on a light purple gradient background. The form is a white rounded rectangle with a shadow. At the top, it has the title 'Teacher Registration' in bold, followed by a note: 'Accounts require administrative approval before activation.' Below this are four input fields: 'Full Name', 'Email Address', 'Password', and 'Confirm Password'. The 'Password' and 'Confirm Password' fields have an eye icon to toggle visibility. At the bottom of the form is a blue 'Create Account' button. Below the button is a link: 'Already registered? [Back to Login](#)'.

Figure 5.1 Teacher Registration



The image shows an 'EduDOC Login' form centered on a light blue gradient background. The form is a white rounded rectangle with a shadow. At the top left of the page, there is a link: '← [Back to Home](#)'. The form has the title 'EduDOC Login' in bold. Below the title are two input fields: 'Username' and 'Password'. At the bottom of the form is a blue 'Login' button. Below the button are two links: 'Forgot Password?' and 'New Teacher? [Register here](#)'.

Figure 5.2 Login Page

5.1.2 Admin Management Module

The Admin Management module allows administrators to control and supervise the overall operation of the EduDOC system. The administrator is responsible for managing users, monitoring system activities, and ensuring the integrity of the platform.

One of the primary responsibilities of the administrator is approving teacher accounts before they are granted full system access. The admin module also allows administrators to manage batches of students and assign teachers to specific batches. This ensures that teachers interact only with students belonging to their designated academic groups.

In addition to user management, the administrator can monitor system statistics and activity logs. These logs provide information about system usage, including the number of users, submissions, and evaluation activities.

Key functions of this module include:

- Teacher approval and verification
- Student and batch management
- Teacher–batch assignment
- Monitoring system usage statistics
- Viewing system activity logs

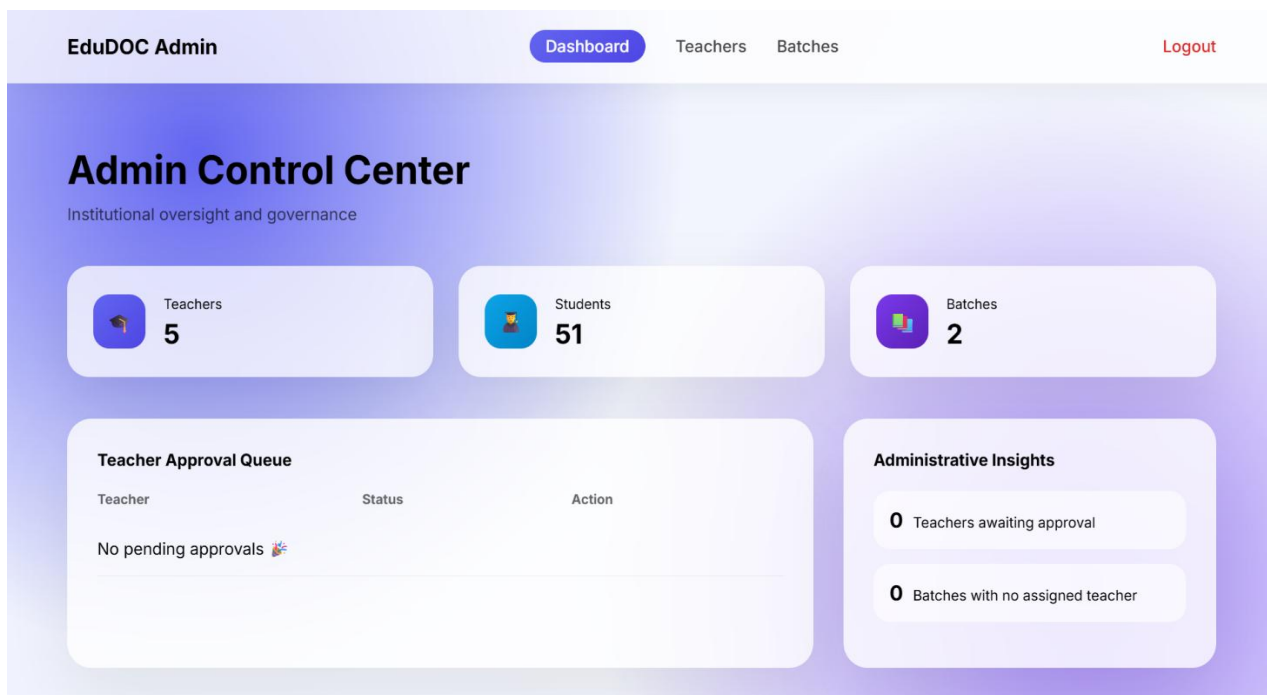


Figure 5.3 Admin Dashboard

5.1.3 Teacher Classroom Management Module

The Teacher Classroom Management module enables instructors to create and manage academic activities within the EduDOC system. In this system, a classroom represents an assignment or evaluation activity where students submit their documents for compliance verification.

Teachers can create classrooms and configure document guidelines that specify the required formatting standards for student submissions. These guidelines may include rules related to margin size, font type, font size, line spacing, and document structure.

The module also allows teachers to define submission deadlines and configure submission limits for each classroom. Teachers can monitor student submissions and access detailed compliance reports generated by the system.

The classroom workspace provides an overview of student submissions, evaluation results, and document analysis reports. Teachers can use this information to assess whether students have followed the specified formatting guidelines.

This module simplifies assignment management and allows instructors to evaluate student submissions efficiently.

Key functions of this module include:

- Classroom creation and management
- Uploading and configuring document guidelines
- Setting submission deadlines
- Monitoring student submissions

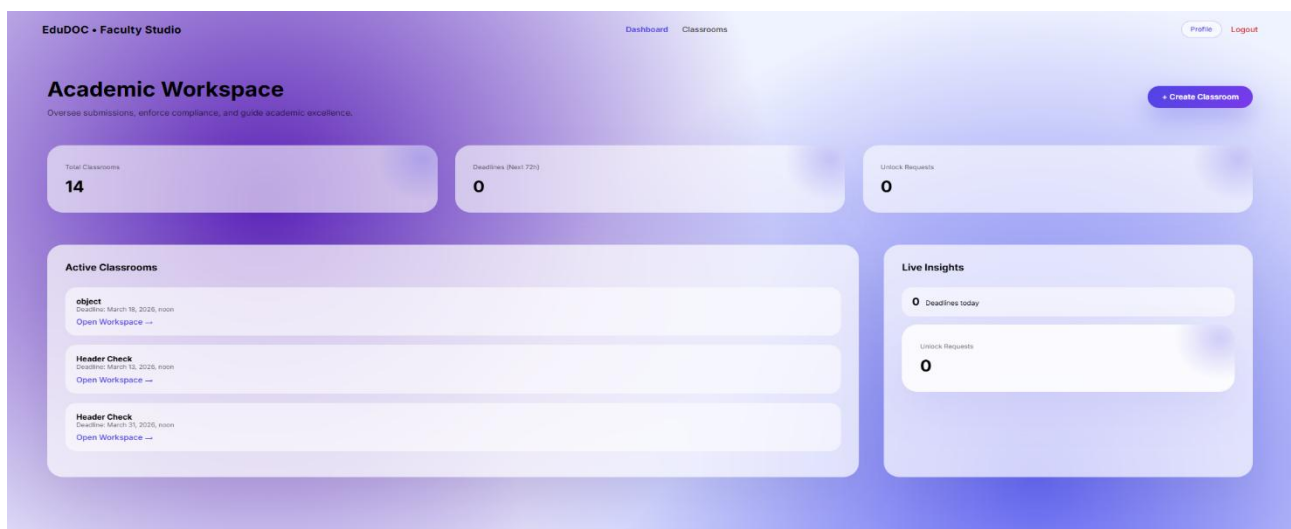


Figure 5.4 Teacher dashboard

5.1.4 Student Submission Module

The Student Submission module allows students to interact with the system by submitting their academic documents for evaluation. Students begin by joining classrooms using a passkey provided by the instructor. Once enrolled in a classroom, they can view assignment details and submission guidelines defined by the teacher.

The primary function of this module is document submission. Students upload their assignments through the submission interface, and the system automatically processes the document for compliance verification.

After evaluation, students can access detailed reports that highlight formatting issues and compliance results. The system may also provide annotated documents where formatting violations are highlighted.

Key functions of this module include:

- Joining classrooms using passkeys
- Uploading assignment documents
- Viewing submission history
- Accessing compliance reports
- Downloading annotated evaluation results

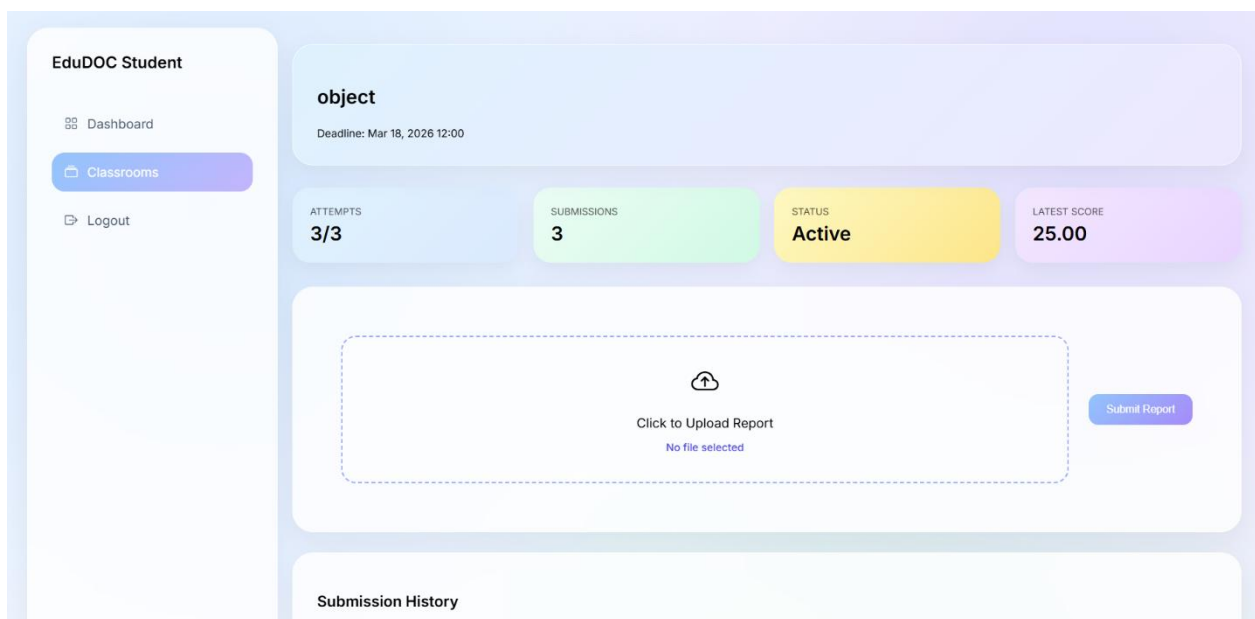


Figure 5.5 Student Submission

5.1.5 Report Generation Module

The Report Generation module compiles the results of the document evaluation process and produces structured reports for both students and teachers. These reports provide a detailed summary of the compliance status of each submitted document.

The system organizes evaluation results into a clear format that highlights formatting rules, detected violations, and compliance outcomes. Reports may also include annotated versions of the submitted document where formatting errors are visually highlighted.

Teachers can review these reports to assess whether student submissions follow the required guidelines. Students can also use the reports to understand their mistakes and improve future submissions.

The report generation module ensures that evaluation results are presented in a transparent and easy-to-understand format.

Key functions of this module include:

- Generation of compliance reports
- Display of formatting violations
- Creation of annotated documents
- Presentation of structured feedback for students and teachers

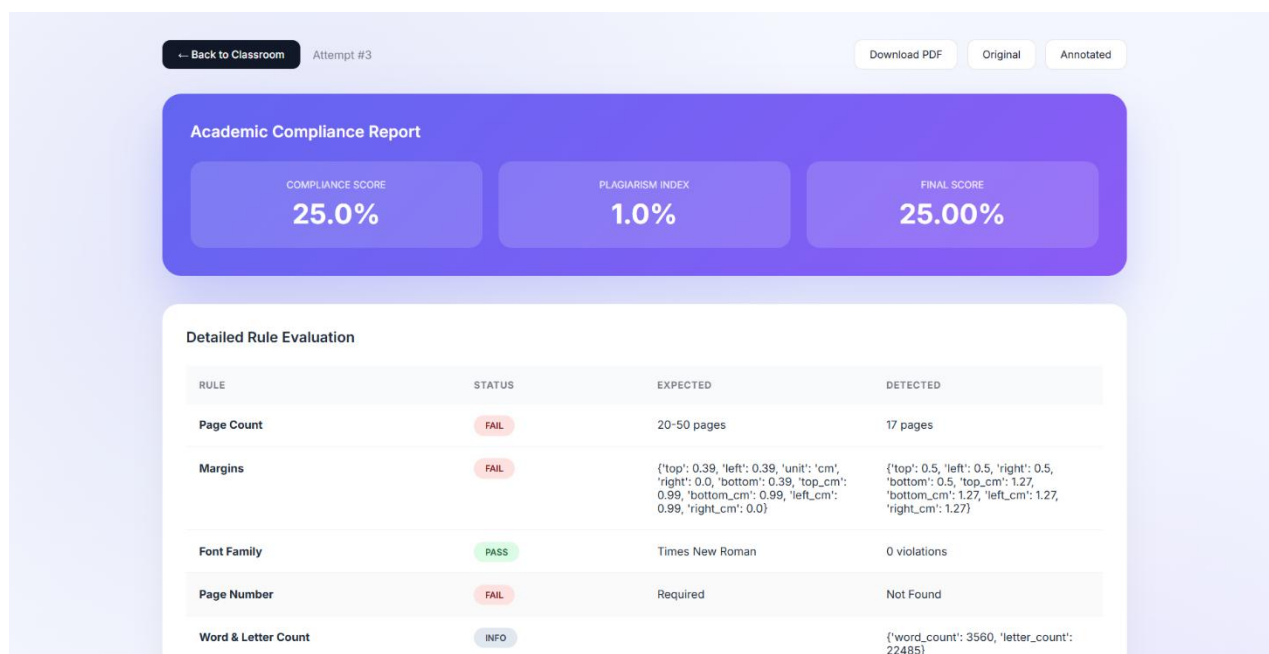


Figure 5.6 Document Evaluation

5.2 Algorithms

The EduDOC system performs automated evaluation of academic documents using a combination of document parsing, rule-based compliance verification, and report generation algorithms. These algorithms enable the system to analyze uploaded documents, extract structural attributes, compare them with predefined guidelines, and produce structured evaluation reports.

The algorithms implemented in the system ensure that document verification is performed consistently and efficiently without requiring manual inspection by instructors.

5.2.1 Document Parsing Algorithm

The first step in the evaluation process involves parsing the uploaded document to extract relevant structural and formatting information. The document parsing algorithm reads the uploaded document file and identifies key elements such as paragraphs, font properties, margins, and spacing values.

The algorithm processes the document sequentially and extracts formatting attributes from each paragraph and text segment. These attributes are then stored temporarily for further analysis.

Algorithm: Document Parsing

1. Start
2. Receive the uploaded document file from the student submission module
3. Open the document using a document processing library
4. Iterate through each paragraph in the document
5. Extract formatting attributes such as:
 - Font name
 - Font size
 - Paragraph spacing
 - Alignment
 - Margins
6. Store extracted attributes in a structured data format
7. Return extracted document attributes for compliance verification
8. End

5.2.2 Formatting Compliance Verification Algorithm

Once the document attributes have been extracted, the system performs **rule-based verification** to determine whether the document follows the formatting guidelines defined by the teacher.

The compliance verification algorithm compares each extracted attribute with the predefined rules stored in the system database. These rules may include parameters such as font type, font size, margin values, paragraph spacing, and heading structure. Each extracted attribute is evaluated individually against the corresponding rule to determine whether it satisfies the specified requirement.

If a mismatch is detected between the document attribute and the predefined rule, the system records the violation and marks the corresponding rule as failed. The evaluation results are stored in a structured format so that the system can generate detailed compliance reports.

After verifying all rules, the algorithm calculates the overall compliance status of the document. The results are then used to generate a compliance report that highlights formatting violations and provides feedback to the user. This rule-based verification approach ensures consistent and objective evaluation of document formatting according to the guidelines configured within the system.

Algorithm: Formatting Compliance Verification

1. Start
2. Receive extracted document attributes from the parsing module
3. Retrieve predefined formatting rules from the classroom configuration
4. For each rule in the rule set:
 - Compare the expected rule value with the extracted document value
5. If values match:
 - Mark rule as PASS
6. If values do not match:
 - Mark rule as FAIL
 - Record the detected violation
7. Store compliance results in the evaluation dataset

8. Return compliance results to the report generation module
9. End

This algorithm ensures that document formatting rules are applied consistently to all submissions.

5.2.3 Spelling Detection Algorithm

The EduDOC system includes a spelling verification mechanism that analyzes textual content within the document to identify potential spelling mistakes. This algorithm contributes to improving the overall quality of academic writing by detecting incorrectly spelled words within the submitted document.

The spelling detection process begins after the textual content is extracted from the uploaded document. The system separates the text into individual words through a tokenization process. Each word is then compared against a predefined dictionary that contains valid words in the language.

If a word is not found in the dictionary, the system marks it as a potential spelling error. These identified words are recorded as informational issues in the evaluation report. The system also highlights such words in the annotated version of the document so that students can easily locate and correct them.

By performing automated spelling verification, the system provides additional feedback regarding document quality. This mechanism supports students in improving the clarity and correctness of their academic writing while preparing properly formatted documents.

Algorithm: Spelling Detection

1. Start
2. Extract textual content from the uploaded document
3. Tokenize the text into individual words
4. For each word in the document:
 - Check whether the word exists in the dictionary
5. If the word is not found in the dictionary:
 - Mark the word as a spelling error
6. Record all detected spelling errors
7. Return spelling analysis results to the evaluation report module

8. End

This process enables the system to highlight spelling mistakes in the annotated document.

5.2.4 Evaluation Report Generation Algorithm

After completing document analysis and compliance verification, the system generates a structured evaluation report summarizing the results of the document evaluation process.

The report generation algorithm compiles all detected formatting violations, spelling issues, and compliance results into a structured format that can be viewed by both teachers and students. The algorithm retrieves the evaluation results produced during the compliance verification stage and organizes them into categorized sections.

Each rule evaluated by the system is listed along with its corresponding status, indicating whether the rule has been satisfied or violated. Detected spelling issues and additional document statistics, such as word count or page count, are also included in the report to provide further insights into document quality.

The generated report is stored in the system database and made accessible through the user dashboards. In addition to the structured report, the system produces an annotated version of the document in which detected issues are highlighted directly within the document. This allows users to easily identify formatting violations and spelling mistakes.

Through this report generation process, the system provides clear and transparent feedback regarding document compliance, enabling students to understand and correct formatting errors while allowing teachers to efficiently review evaluation results.

Algorithm: Evaluation Report Generation

1. Start
2. Receive compliance verification results
3. Receive spelling analysis results
4. Compile evaluation data into structured format
5. Generate compliance summary including:
 - Passed rules
 - Failed rules
 - Detected formatting violations

6. Generate annotated document highlighting issues
7. Store evaluation results in the system database
8. Display evaluation report on user dashboard
9. End

The report generation process ensures that evaluation results are presented clearly, allowing both instructors and students to review the analysis easily.

These algorithms collectively enable the EduDOC system to perform automated document evaluation, ensuring consistent application of formatting rules and providing meaningful feedback to students regarding document compliance and quality.

5.3 System Architecture

The EduDOC system is designed using a web-based client–server architecture that enables users to access the platform through a standard web browser while the core processing and data management operations are handled by the server. This architecture ensures scalability, maintainability, and efficient handling of document evaluation tasks.

The architecture follows a three-layered structure, consisting of the Presentation Layer, Application Layer, and Data Layer. Each layer is responsible for specific system functionalities and interacts with other layers to process user requests and deliver responses.

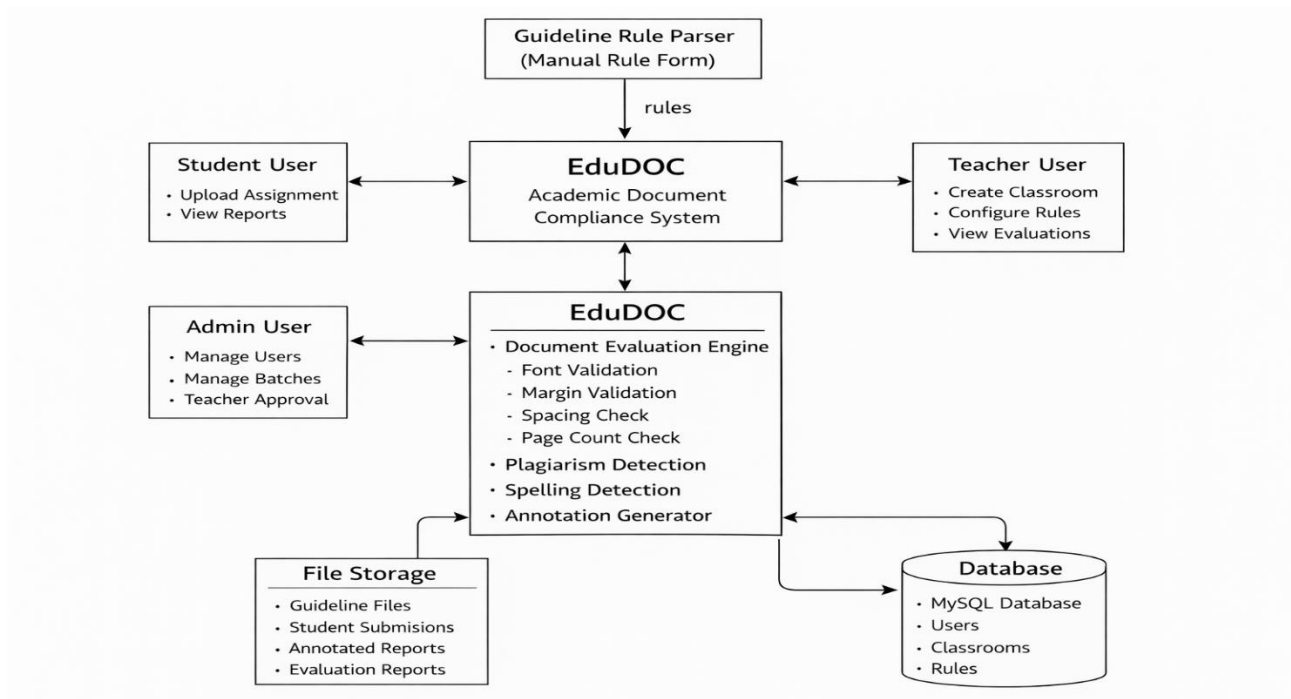


Figure 5.7: System Architecture

5.3.1 Presentation Layer

The Presentation Layer represents the user interface of the EduDOC system. This layer allows users such as administrators, teachers, and students to interact with the system through web-based dashboards and interfaces.

Users access the system through a web browser where different dashboards are displayed according to the role assigned to the user. The presentation layer includes components such as login pages, dashboards, classroom interfaces, document upload forms, and evaluation report views.

The interface is designed to provide a clear and structured user experience so that users can easily perform tasks such as registering accounts, submitting documents, creating classrooms, and reviewing evaluation results.

Key features of the Presentation Layer include:

- User login and authentication interface
- Admin dashboard for system monitoring
- Teacher dashboard for classroom management
- Student dashboard for document submission
- Compliance report viewing interface
- Document upload and evaluation interface

This layer communicates with the application layer to send user requests and display the results returned by the system.

5.3.2 Application Layer

The Application Layer forms the core processing component of the EduDOC system. This layer contains the business logic that processes user requests, performs document evaluation, and manages system operations.

The application layer is responsible for coordinating interactions between different modules of the system. When a user performs an action such as uploading a document or creating a classroom, the application layer processes the request and executes the appropriate operations.

Several key modules operate within this layer, including:

- Authentication and User Management Module – Handles user authentication and role-based access control.

- Admin Management Module – Manages teacher approvals, student creation, and system supervision.
- Teacher Classroom Management Module – Allows teachers to create classrooms and define submission guidelines.
- Student Submission Module – Enables students to upload academic documents for evaluation.
- Document Evaluation Module – Analyzes uploaded documents and verifies compliance with formatting rules.
- Report Generation Module – Generates structured compliance reports and annotated documents.

The application layer also processes guideline rules defined by teachers and applies them during document evaluation. The document evaluation engine extracts formatting attributes from uploaded documents and compares them with predefined rules to detect violations.

5.3.3 Data Layer

The Data Layer is responsible for storing and managing all system data. This layer uses a relational database management system to maintain records related to users, classrooms, submissions, guidelines, and evaluation results.

The database ensures that information is stored securely and can be retrieved efficiently when required by the application layer. The system maintains several key data entities including user accounts, classroom configurations, document submissions, compliance rules, and generated evaluation reports.

The database structure is designed to support relationships between different entities. For example, students are associated with specific classrooms, classrooms are linked to teachers, and evaluation reports are connected to individual document submissions.

The Data Layer provides the following functionalities:

- Storage of user account information
- Management of classroom and assignment data
- Storage of document submissions
- Storage of evaluation results and reports

- Retrieval of system data for analysis and reporting

By separating data storage from application logic, the architecture ensures improved data security and easier system maintenance.

5.3.4 Workflow of the System

The interaction between the three architectural layers occurs through a structured workflow. When a user interacts with the system, the request is first received by the presentation layer. The request is then forwarded to the application layer, where the necessary processing is performed.

For example, when a student uploads a document, the system performs the following steps:

1. The student uploads the document through the web interface.
2. The presentation layer sends the request to the application layer.
3. The application layer processes the uploaded document.
4. The document evaluation module extracts formatting attributes.
5. Extracted attributes are compared with predefined guideline rules.
6. Evaluation results are generated and stored in the database.
7. The report generation module creates a structured compliance report.
8. The results are returned to the presentation layer and displayed to the user.

This workflow ensures that document evaluation is performed efficiently and that results are delivered to users in a structured format.

5.3.5 Advantages of the Architecture

The architecture used in the EduDOC system offers several advantages that improve the performance and usability of the platform.

First, the layered structure allows the system to be modular and scalable, making it easier to add new features or update existing modules without affecting the entire system.

Second, the separation of presentation, application, and data layers improves system maintainability and reliability. Developers can modify individual components independently without disrupting other parts of the system.

Third, the client–server model enables remote accessibility, allowing users to access the system from different locations using a web browser.

Finally, the architecture supports efficient processing of document evaluation tasks, ensuring that large numbers of submissions can be analyzed automatically without requiring manual inspection.

5.4 Case Study

To demonstrate the practical application and effectiveness of the EduDOC system, a case study was conducted to simulate the complete workflow of academic document submission and evaluation within a classroom environment. The case study illustrates how different users interact with the system and how automated document compliance verification assists instructors in evaluating academic submissions.

The case study involves three primary roles in the system: Administrator, Teacher, and Student. Each role performs specific actions that contribute to the overall document evaluation process.

5.4.1 Teacher Registration and Approval

The process begins with a teacher registering on the EduDOC platform through the teacher registration interface. During registration, the teacher provides the necessary information such as name, email address, and password.

After registration, the teacher account remains inactive until it is approved by the system administrator. The administrator reviews the teacher's registration request through the admin dashboard and verifies the information before granting approval.

Once the account is approved, the teacher is able to log into the system and access the teacher dashboard.

This approval mechanism ensures that only verified instructors are allowed to create classrooms and manage academic activities within the platform.

5.4.2 Student Account Creation

Unlike teachers, students do not register themselves in the system. Instead, student accounts are created by the administrator. The administrator can add students individually or upload student data in bulk using a spreadsheet or CSV file.

After the accounts are created, students receive login credentials that allow them to access the system. Students can then log in and view the classrooms they are eligible to join.

This controlled account creation process helps maintain accurate student records and ensures that only authorized students can access the platform

5.4.3 Classroom Creation by Teacher

Once logged into the system, the teacher can create a classroom that represents an academic activity such as an assignment or project submission.

During classroom creation, the teacher specifies several parameters including:

- Classroom name
- Submission deadline
- Maximum marks
- Formatting guidelines for the document

The teacher may also upload a guideline document that contains formatting instructions for students. These guidelines are later used by the system to evaluate whether submitted documents comply with the required formatting standards.

After the classroom is created, the system generates a **classroom passkey**. This passkey is shared with students so they can join the classroom.

5.4.4 Student Joining Classroom and Submitting Document

Students log into the system and join the classroom using the passkey provided by the teacher. After joining the classroom, students can view the assignment instructions and submission deadline.

Students prepare their academic documents according to the guidelines provided by the teacher and upload the document through the submission interface. The uploaded document is then stored in the system and sent to the document evaluation module for automated analysis. The system also tracks the number of submission attempts and prevents further submissions if the allowed attempt limit is reached.

5.4.5 Automated Document Evaluation

After a document is uploaded, the EduDOC system automatically initiates the evaluation process. The document evaluation module extracts formatting attributes from the uploaded file and compares them with the predefined rules specified by the teacher.

The evaluation process checks several formatting elements including:

- Font type
- Font size

- Paragraph spacing
- Margin size
- Heading structure
- Page / Word count
- Spelling consistency

If any formatting rule is violated, the system records the issue and marks the corresponding rule as failed.

5.4.6 Report Generation and Feedback

Once the evaluation process is completed, the system generates a detailed compliance report. The report displays the status of each formatting rule and highlights any detected violations. The system may also generate an annotated version of the submitted document where formatting errors and spelling mistakes are visually highlighted. This allows students to quickly identify the sections of the document that require correction.

Teachers can view the evaluation reports through the classroom workspace, while students can access their individual reports through the student dashboard. The structured feedback provided by the system helps students improve their document preparation skills.

5.4.7 Outcome of the Case Study

The case study demonstrates that the EduDOC system successfully automates the evaluation of academic documents by analyzing formatting rules and generating structured reports. The automated workflow significantly reduces the time required for instructors to verify document compliance while ensuring consistent application of formatting guidelines.

The system also benefits students by providing clear and immediate feedback regarding formatting issues in their documents. This feedback enables students to correct errors and improve the quality of their submissions.

Overall, the case study highlights the effectiveness of the EduDOC platform in streamlining the document submission and evaluation process within academic environments.

CHAPTER 6

IMPLEMENTATION DETAILS

6.1 Implementation Procedures

The implementation of the EduDOC system was carried out using a structured and incremental development approach to ensure that each component of the system was properly developed, tested, and integrated. The development process involved designing the database schema, implementing backend application logic, developing user interfaces, and integrating the document evaluation engine responsible for automated compliance verification.

The system was implemented as a web-based application using modern web development technologies. The backend logic was implemented using a server-side web framework, while the database was used to store user accounts, classroom data, document submissions, evaluation results, and system logs. The overall system architecture follows a modular design, allowing different functional modules to operate independently while remaining integrated within the main platform.

System Environment Setup

The first stage of implementation involved configuring the development environment required to build and run the application. This included installing the necessary programming tools, setting up the web framework, and configuring the database server.

The database schema was created according to the system design specifications. Tables were defined to store user information, classroom details, student enrollments, document submissions, and evaluation reports. After configuring the database, the project structure was initialized and the application modules were organized within the project directory to support modular development.

User Authentication Implementation

The next stage involved implementing the authentication and access control system that enables secure access to the platform. A role-based authentication mechanism was designed to support three types of users: administrator, teacher, and student.

Teachers can register through the system interface; however, their accounts remain inactive until approved by the administrator. This approval mechanism ensures that only verified instructors are allowed to create classrooms and manage academic activities. Student accounts are created and

managed by the administrator and stored in the database, allowing controlled enrollment and access to the platform.

Admin Management Implementation

The administrator module was implemented to provide centralized management of system operations. The administrator dashboard allows administrators to review and approve teacher registration requests, manage student accounts, and monitor overall system activity.

Batch management functionality was implemented to organize students into academic groups. These batches allow teachers to interact only with students assigned to their respective academic groups. Administrative access controls were integrated with the authentication system to ensure that only authorized administrators can access and manage system-level features.

Teacher Classroom Management Implementation

The teacher module was developed to allow instructors to create and manage classrooms within the system. Each classroom functions as an assignment workspace where students can submit their academic documents for evaluation.

Teachers can configure classroom parameters such as submission deadlines, maximum submission attempts, and document formatting rules. In addition, teachers can upload guideline documents that define formatting requirements for student submissions.

When a classroom is created, the system generates a unique classroom passkey that allows students to join the classroom. The classroom workspace provides teachers with access to submission records, evaluation reports, and compliance summaries for submitted documents.

Student Submission Implementation

The student module enables students to upload academic documents for automated evaluation. Students join classrooms by entering the passkey provided by the teacher.

The document upload functionality supports structured document formats and securely stores uploaded files on the server. Each uploaded document is linked to a corresponding submission record within the database. The system also monitors submission attempts and enforces restrictions such as maximum attempts and submission deadlines based on the rules configured by the teacher.

Document Evaluation Engine Implementation

The document evaluation engine was implemented as the core processing component of the EduDOC system. When a student uploads a document, the evaluation engine processes the file and extracts structural formatting attributes such as font type, font size, margin settings, paragraph spacing, and heading structure.

These extracted attributes are compared against predefined formatting rules configured within the classroom settings. The system determines whether each rule is satisfied or violated and records the results for evaluation reporting.

Additional checks such as spelling detection and word count analysis were integrated into the evaluation engine to provide further feedback regarding the quality and completeness of the submitted document.

Report Generation Implementation

After the evaluation process is completed, the system generates structured compliance reports summarizing the results of the document analysis. These reports provide information regarding formatting compliance, detected rule violations, and evaluation outcomes.

The system also generates annotated versions of submitted documents, highlighting formatting violations directly within the document. This feature allows students to quickly identify and correct formatting errors.

All evaluation reports are stored in the database and can be accessed by both students and teachers through their respective dashboards.

Through these implementation procedures, the EduDOC system successfully integrates user management, classroom management, document evaluation, and automated report generation into a unified platform for academic document compliance verification.

6.2 Testing Methods and Results

Testing is a critical phase in software development to ensure that the system operates correctly and produces reliable results. Several testing strategies were applied during the development of the EduDOC system to validate system functionality, detect potential issues, and verify the accuracy of the document evaluation engine.

Unit Testing

Unit testing was conducted to verify the functionality of individual modules within the system. Each module was tested independently to ensure that it performs its intended operations correctly.

Modules tested during unit testing included:

- User authentication and login validation
- Classroom creation and management
- Student enrollment and submission tracking
- Document parsing and formatting attribute extraction

These tests ensured that each component of the system operates correctly before integration with other modules.

Integration Testing

Integration testing was performed to verify that different system modules interact correctly with one another. This stage focused on validating the flow of data between the authentication system, classroom management module, submission processing module, and the document evaluation engine.

For example, integration testing verified that when a student uploads a document, the system correctly transfers the file to the evaluation engine, processes the document, and generates a compliance report.

Functional Testing

Functional testing was conducted to confirm that the system performs according to the specified system requirements. A set of test scenarios was designed to simulate real-world system usage.

These scenarios included:

- Teacher registration and administrator approval
- Classroom creation and guideline configuration
- Student classroom enrollment using passkeys
- Document upload and automated evaluation
- Generation and display of compliance reports

User Interface Testing

User interface testing was performed to ensure that the system dashboards and forms provide a clear and user-friendly experience. This testing verified that users can easily navigate the platform and perform tasks without encountering usability issues.

The testing also ensured that the system properly displays error messages, alerts, and feedback notifications when users perform invalid or incomplete actions.

Testing Results

The testing results demonstrated that the EduDOC system successfully detects formatting violations and generates accurate compliance reports. Documents that followed the defined formatting guidelines were correctly identified as compliant, while documents containing formatting errors were properly flagged by the system.

The automated evaluation process significantly reduced the time required for document formatting verification compared to manual review. In addition, the system provides clear feedback through structured reports and annotated documents, enabling students to identify and correct formatting issues efficiently.

Overall, the testing process confirmed that the EduDOC system operates reliably and provides an effective platform for automated academic document compliance verification.

CHAPTER 7

ADVANTAGES AND DISADVANTAGES

7.1 Advantages

The EduDOC system provides several advantages for educational institutions by automating the process of academic document compliance verification and evaluation. By integrating document analysis technologies with a web-based submission platform, the system simplifies the management of assignments and improves the efficiency of document evaluation.

One of the primary advantages of the system is the automation of document formatting verification. In traditional academic environments, instructors are required to manually inspect each document to ensure that it follows formatting guidelines such as font style, font size, margins, and line spacing. This process can be repetitive and time-consuming, especially when dealing with large numbers of submissions. The EduDOC system automatically analyzes these formatting attributes, allowing instructors to focus more on evaluating the conceptual content of student work.

Another significant advantage is the reduction of manual workload for teachers. Since the system automatically evaluates document formatting and generates compliance reports, instructors do not need to manually check every formatting rule. This improves the efficiency of the evaluation process and allows instructors to manage assignments more effectively.

The system also promotes consistency and fairness in document evaluation. When document verification is performed manually, there is a possibility of variation in the interpretation of formatting guidelines among different instructors. The automated evaluation process ensures that the same rules are applied consistently to all submitted documents, reducing subjectivity in the evaluation process.

EduDOC also provides structured and detailed feedback to students. After document evaluation, the system generates reports that clearly indicate which formatting rules have been satisfied and which rules have been violated. Students can use this feedback to identify errors in their documents and improve their formatting practices in future submissions.

Another important advantage is the role-based access control system implemented within the platform. Administrators, teachers, and students interact with the system through dedicated interfaces tailored to their responsibilities. This structured access control enhances system security and ensures that users can access only the functionalities relevant to their roles.

The system also improves document management and record keeping. All submissions, evaluation results, and user activities are stored in the database, allowing instructors and administrators to track submission histories and monitor academic activities efficiently.

Furthermore, the web-based nature of the system provides remote accessibility, enabling users to access the platform from different locations using standard web browsers. This makes the system suitable for modern digital learning environments.

7.2 Disadvantages

Although the EduDOC system offers several benefits, certain limitations and challenges are associated with its implementation. Understanding these limitations is important for improving the system in future developments.

One limitation of the system is its dependence on structured document formats. The document evaluation engine is designed to analyze documents prepared in specific formats supported by the document processing library. If a document is uploaded in an unsupported format or contains unusual formatting structures, the system may not be able to extract all required formatting attributes accurately.

Another potential limitation is that the system focuses primarily on formatting compliance rather than deep content evaluation. While the system can analyze formatting rules and detect spelling issues, it does not perform comprehensive semantic analysis of the document content. Therefore, instructors must still review the intellectual quality and originality of the academic work.

The accuracy of automated evaluation may also depend on the quality of the predefined rules configured by the teacher. If formatting guidelines are incorrectly defined or incomplete, the evaluation results may not fully reflect the intended requirements.

The system also requires initial configuration and administrative setup before it can be used effectively. Administrators must create student accounts, approve teachers, and configure batches before users can interact with the system. This setup process may require additional time during the initial deployment stage.

Another limitation is related to system dependency on internet connectivity. Since the platform operates as a web-based application, users must have internet access to upload documents, view reports, or interact with the system.

Despite these limitations, most of the challenges can be addressed through system enhancements, improved rule configuration, and integration of advanced document analysis technologies in future versions of the platform.

CHAPTER 8

RESULT

8.1 Analysis

The performance of the EduDOC system was analyzed by examining its ability to process academic document submissions and verify compliance with predefined formatting guidelines. The analysis focused on evaluating how effectively the system extracts document attributes, applies formatting rules, and generates structured evaluation reports.

During testing, a variety of sample academic documents were submitted through the student interface. These documents were intentionally prepared with different formatting configurations to evaluate the accuracy of the document evaluation engine.

Some documents strictly followed the formatting guidelines defined by the teacher, while others included formatting violations such as incorrect font sizes, inconsistent margins, improper line spacing, and missing headings.

When documents were uploaded to the system, the document evaluation module successfully parsed the document structure and extracted relevant formatting attributes. These attributes included font type, font size, paragraph alignment, spacing, margin settings, and heading structures. The extracted attributes were then compared with the rule set configured for the classroom.

The analysis revealed that the rule-based evaluation engine accurately detected formatting violations in documents that did not comply with the defined guidelines. Each violation was recorded and presented in the compliance report, allowing both instructors and students to identify specific formatting issues.

Another important aspect analyzed was the system's ability to generate annotated documents. The annotated output files clearly highlighted sections of the document where formatting rules were violated. This feature provided visual feedback that helped students quickly understand the corrections required in their documents.

The system was also evaluated for its ability to manage multiple document submissions efficiently. The evaluation engine processed each uploaded document automatically and produced compliance reports without requiring manual intervention from the instructor.

This demonstrates that the system can handle document evaluation tasks efficiently even when multiple submissions are processed.

8.2 Results

The implementation of the EduDOC system was evaluated by testing its ability to process academic document submissions and verify formatting compliance according to predefined guidelines. Several sample documents were submitted through the student interface and analyzed by the document evaluation engine.

During testing, documents with different formatting characteristics were used to evaluate the accuracy and effectiveness of the compliance verification process. Some documents were intentionally prepared with formatting errors, while others followed the required guidelines correctly. This approach allowed the system to be tested under different scenarios.

When a student uploaded a document, the system successfully extracted formatting attributes such as font style, font size, paragraph spacing, margins, heading structures, and word count. These attributes were then compared with the rules defined by the teacher in the classroom configuration.

The results showed that the system correctly identified formatting violations in documents that did not comply with the required guidelines. For instance, documents containing incorrect font sizes, improper spacing, or inconsistent margins were detected by the evaluation engine and marked as non-compliant.

Documents that followed all formatting requirements were successfully classified as compliant. The system generated structured evaluation reports that clearly indicated which formatting rules were satisfied and which rules were violated.

Another important result observed during testing was the effectiveness of the annotated document generation feature. In the annotated output files, formatting violations and spelling issues were highlighted within the document itself. This allowed users to visually identify problematic sections and understand the required corrections. Overall, the results indicate that the EduDOC system successfully performs automated document compliance verification and provides useful feedback to both students and instructors.

8.3 Discussion

The results obtained from the testing process demonstrate that the EduDOC system effectively addresses several challenges associated with manual document evaluation in academic environments.

One of the most significant improvements observed is the reduction in the time required for formatting verification. In traditional evaluation processes, instructors must manually review each document to check formatting compliance. This process can be time-consuming and repetitive,

particularly when evaluating a large number of submissions. By automating formatting verification, the EduDOC system significantly reduces the effort required from instructors.

Another important advantage is the consistency of evaluation results. Since the system applies predefined formatting rules automatically, all documents are evaluated using the same criteria. This eliminates variations that may occur when multiple instructors interpret formatting guidelines differently.

The structured evaluation reports generated by the system also improve communication between teachers and students. Instead of receiving vague feedback about formatting errors, students receive detailed reports that clearly explain which rules were violated and where corrections are needed.

The annotated document feature further enhances the usefulness of the system by visually highlighting formatting violations. This visual feedback helps students quickly understand the issues in their documents and correct them more effectively.

However, the discussion also reveals certain limitations of the current system. The evaluation engine primarily focuses on formatting compliance and basic textual analysis. While these features are valuable, deeper semantic analysis of document content is not performed. Therefore, instructors must still review the conceptual quality and originality of the submitted work.

Despite these limitations, the EduDOC system demonstrates significant potential for improving the efficiency and reliability of academic document evaluation. By automating formatting verification and generating structured feedback, the system supports both instructors and students in maintaining high standards for academic document preparation.

CHAPTER 9

CONCLUSION

9.1 Summary

The increasing use of digital platforms for academic activities has created a need for efficient systems that can assist educators in managing and evaluating academic documents. In many educational institutions, students are required to submit assignments, project reports, and research documents that follow specific formatting and structural guidelines.

Ensuring that these guidelines are properly followed is an important aspect of academic evaluation. However, verifying formatting compliance manually can be a repetitive and time-consuming task for instructors, particularly when dealing with a large number of student submissions.

The EduDOC system, titled “*An Automated Framework for Academic Document Compliance and Evaluation*,” was developed to address this challenge by introducing an automated approach to document compliance verification. The system provides a web-based platform that enables administrators, teachers, and students to interact with a structured academic document evaluation environment.

Through this platform, teachers can define formatting guidelines for assignments, students can submit their documents electronically, and the system can automatically analyze these documents to verify compliance with predefined rules.

The development of the system involved designing a modular architecture that integrates several functional components, including authentication and user management, classroom management, document submission, automated document evaluation, and report generation. The platform ensures secure access through role-based authentication, where administrators manage system users, teachers configure classroom guidelines, and students submit their documents for evaluation.

A key component of the system is the document evaluation engine, which automatically analyzes uploaded documents and extracts formatting attributes such as font style, font size, paragraph spacing, margin settings, and heading structures. These attributes are compared with the formatting guidelines defined by the teacher within the classroom configuration.

The system then generates a detailed compliance report indicating whether each rule has been satisfied or violated. In addition to formatting verification, the system also performs supplementary

checks such as word count analysis and spelling detection to provide additional insights into the quality of the submitted document.

The implementation of EduDOC demonstrates how automation can significantly improve the efficiency and reliability of academic document evaluation. By automatically verifying formatting rules, the system reduces the need for manual document inspection and ensures that evaluation criteria are applied consistently across all submissions. Teachers can review structured compliance reports instead of manually checking each document, while students receive clear feedback that helps them correct formatting errors and improve their document preparation practices.

Testing and evaluation of the system confirmed that the document evaluation engine accurately detects formatting inconsistencies and generates meaningful compliance reports. The system successfully processed documents with different formatting characteristics and identified violations where formatting guidelines were not followed. The structured feedback generated by the system allows students to easily identify and correct formatting issues.

Overall, the EduDOC system demonstrates the effectiveness of integrating automated document analysis techniques into academic environments. The platform streamlines the document submission and evaluation process, enhances the consistency of formatting verification, and supports both instructors and students in maintaining high standards for academic document preparation.

9.2 Limitations

Although the EduDOC system provides an effective solution for automated document compliance verification, certain limitations exist within the current implementation.

One limitation of the system is its dependence on specific document formats supported by the document processing libraries used during implementation. If documents contain unusual formatting structures or are submitted in unsupported formats, the system may not be able to extract all formatting attributes accurately. Variations in document creation tools and formatting styles may also lead to minor inconsistencies in the extracted document properties.

Another limitation is that the system primarily focuses on formatting compliance and basic textual analysis. While the system can detect formatting inconsistencies and spelling errors, it does not perform advanced semantic analysis of the document content. As a result, the system cannot fully evaluate the conceptual quality, argument strength, or contextual correctness of the written material.

The effectiveness of the evaluation process also depends on the accuracy of the formatting rules configured by the teacher. If the guidelines are incomplete, ambiguous, or incorrectly defined, the

evaluation results may not fully reflect the intended formatting standards. In such cases, the automated verification process may produce incomplete or partially accurate compliance reports.

In addition, plagiarism detection in the current system is limited to similarity-based comparison techniques and may not detect highly sophisticated forms of paraphrased plagiarism or semantic content reuse.

Despite these limitations, the system provides a strong foundation for automated academic document evaluation and can be further improved through future enhancements and integration of advanced document analysis techniques.

9.3 Future Scope

The EduDOC system can be expanded and enhanced in several ways to improve its capabilities and extend its functionality in academic environments.

One potential enhancement is the integration of advanced artificial intelligence techniques for deeper document analysis. Machine learning and natural language processing technologies could be used to analyze writing quality, detect grammatical issues, and evaluate the coherence and structure of academic documents.

Another future improvement could involve integrating plagiarism detection mechanisms that compare submitted documents with external sources and institutional databases. This would allow the system to verify both formatting compliance and content originality within a single platform.

The system could also be enhanced to support additional document formats and more complex formatting rules. This would allow the platform to be used across a wider range of academic programs and document types. In addition, mobile-friendly interfaces or dedicated mobile applications could be developed to allow users to access the platform more conveniently from smartphones or tablets.

Integration with institutional learning management systems could also improve the usability of the platform by allowing automatic synchronization of assignments, submissions, and evaluation results. These future developments would further strengthen the capabilities of the EduDOC system and enhance its usefulness in modern digital learning environments.

9.4 Applications

The EduDOC system has several practical applications within academic and educational environments. The platform can be used by educational institutions to automate the process of

verifying document formatting guidelines for assignments, project reports, and research submissions.

One major application of the system is in universities and colleges, where instructors must evaluate large numbers of student submissions. The automated evaluation engine helps reduce the time required for formatting verification while ensuring consistent application of academic guidelines.

The system can also be used in academic research environments, where researchers must prepare documents according to strict formatting standards defined by journals or institutions. Automated formatting verification can assist researchers in ensuring that their documents comply with these requirements. Another potential application is in **training and educational workshops**, where students are learning academic writing and document preparation techniques. The structured feedback provided by the system helps students understand formatting rules and improve their document preparation skills.

Additionally, the system can be adapted for use in **corporate or professional environments** where technical documentation must follow predefined formatting standards. Automated document verification can assist organizations in maintaining consistent documentation practices.

CHAPTER 10

APPENDICES

```
class DocumentEvaluator:
```

```
    PARAGRAPHS_PER_PAGE = 30
```

```
    # =====
```

```
    # INIT
```

```
    # =====
```

```
    def __init__(self, file_path, rules=None, plagiarism_enabled=True):
```

```
        self.file_path = file_path
```

```
        self.rules = rules or {}
```

```
        self.plagiarism_enabled = plagiarism_enabled
```

```
        self.issues = []
```

```
        self.total_checks = 0
```

```
        self.failed_checks = 0
```

```
        self.doc = None
```

```
    # =====
```

```
    # SAFE HELPER (Required by extended rules)
```

```
    # =====
```

```
    def _add_issue(self, rule, message):
```

```
        self.total_checks += 1
```

```
        self.failed_checks += 1
```

```
        self.issues.append({
            "rule": rule,
            "expected": "Valid",
            "found": message,
            "status": "FAIL"
        })
```

```
def evaluate(self):  
    path = Path(self.file_path)  
  
    if not path.exists():  
        return self._empty_result("File not found")  
  
    if path.suffix.lower() != ".docx":  
        return self._empty_result("Unsupported format")  
  
    try:  
        self.doc = Document(self.file_path)  
    except Exception:  
        logger.exception("Failed to open document")  
        return self._empty_result("Unable to read document")  
  
    # Core checks  
    self.check_page_count()  
    self.check_margins()  
    self.check_font_family()  
    self.check_section_order()  
    self.check_page_numbering()  
    self.check_word_and_letter_count()  
    self.check_reference_format()  
  
    # Extended checks  
    self.evaluate_header_footer()  
  
    self._check_font_sizes()  
    self._check_chapter_numbering()  
    self._check_subheading_numbering()  
    self._check_grammar()
```

```
class InternetSearchService:
```

```
    @staticmethod
```

```
    def search_sources(text, max_results=5):  
        """  
        Search internet for matching sources.  
        Returns list of {title, url}  
        """  
  
        # Use first 40 words as search query  
        query = " ".join(text.split()[:40])  
  
        results_list = []  
  
        try:  
            with DDGS() as ddgs:  
                results = ddgs.text(query, max_results=max_results)  
  
                for r in results:  
                    results_list.append({  
                        "title": r["title"],  
                        "url": r["href"]  
                    })  
  
        except Exception as e:  
            print("Internet search failed:", e)  
  
        return results_list
```



```

class ClassroomFinalizer:

    @staticmethod
    def finalize(classroom):
        """
        Finalizes marks for all students in a classroom
        using their LAST attempt only.

        Runs ONLY if deadline has passed.
        Recalculates automatically if a newer attempt exists.
        """

        # Ensure deadline passed
        if not classroom.deadline or timezone.now() <= classroom.deadline:
            return

        student_ids = (
            Submission.objects
            .filter(classroom=classroom)
            .values_list("student_id", flat=True)
            .distinct()
        )

        for student_id in student_ids:

            # Get latest attempt
            last_submission = (
                Submission.objects
                .filter(classroom=classroom, student_id=student_id)
                .order_by("-attempt_no")
                .first()
            )

```

```

def check_page_count(self):

    rules = self.rules.get("page_rules", {})

    min_pages = rules.get("min_pages")
    max_pages = rules.get("max_pages")

    if not min_pages and not max_pages:
        return

    approx_pages = max(
        1,
        round(len(self.doc.paragraphs) / self.PARAGRAPHS_PER_PAGE)
    )

    passed = True

    if min_pages and approx_pages < min_pages:
        passed = False

    if max_pages and approx_pages > max_pages:
        passed = False

    self._record(
        "Page Count",
        f"{min_pages}-{max_pages} pages",
        f"{approx_pages} pages",
        passed
    )

```

```
def generate_annotated_file(self, output_path):

    if not self.doc:
        self.doc = Document(self.file_path)

    doc = self.doc

    # =====
    # 1 INSERT COVER REVIEW SUMMARY
    # =====
    summary = doc.add_paragraph()
    summary.add_run("EduDOC Academic Compliance Review\n").bold = True
    summary.add_run(f"\nCompliance Score : {self._calculate_compliance()}\n")
    summary.add_run(f"Plagiarism Index : {self._plagiarism_stub()}\n")
    summary.add_run(f"Total Violations : {self.failed_checks}\n\n")

    summary.add_run("Major Issues:\n").bold = True

    for issue in self.issues:
        if issue.get("status") == "FAIL":
            summary.add_run(f"• {issue['rule']}\n")

    doc.add_page_break()
```

Link: <https://edudoc-production.up.railway.app>

CHAPTER 11

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